



Machine learning drives a new paradigm in inorganic solid-state electrolytes research

Xiang Yin^{a,b,c}, Jun Ma^{a,b,c}, Shu Zhang^{a,b,c,*}, Guanglei Cui^{a,b,c,*}

^a Qingdao Industrial Energy Storage Research Institute, Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, Qingdao 266101, China

^b Shandong Energy Institute, Qingdao 266101, China

^c Qingdao New Energy Shandong Laboratory, Qingdao 266101, China

ARTICLE INFO

Keywords:

Machine learning
Solid-state electrolyte
High-throughput screening
Property prediction
Inverse design
All-solid-state batteries

ABSTRACT

The development of next-generation solid-state batteries (SSBs) relies heavily on the discovery of high-performance inorganic solid-state electrolytes (SSEs). However, conventional experimental and computational methods are often time-consuming and inefficient in exploring the vast compositional and structural space. Machine learning (ML) has emerged as a powerful and transformative tool, driving a paradigm shift in SSE research. This review systematically summarizes recent advances in ML applications for inorganic SSEs, focusing on its role in predicting and optimizing key properties such as thermodynamic stability, electronic transport, electrochemical stability windows, ionic conductivity, migration barriers, phonon dynamics, mechanical strength, and interfacial stability. By enabling high-throughput screening, establishing structure-property relationships, and guiding inverse design, ML significantly accelerates the discovery and rational design of promising SSE candidates. Perspectives on the challenges such as data quality, model interpretability, and multi-property trade-offs are supplied. Finally, we recommend that future efforts prioritize the development of standardized databases, interpretable and physics-informed ML models, and closed-loop frameworks integrating computation, synthesis, and validation. This review will guide future directions to advance SSE technologies toward practical application in high-energy, safe solid-state batteries.

1. Introduction

Conventional lithium-ion batteries (LIBs) utilizing liquid electrolytes are prone to fire and even explosion risks due to the inherent flammability of organic solvents, while their energy density is constrained as graphite anodes approach fundamental limits [1,2]. To achieve higher energy density, lithium metal stands out as a promising anode material, offering an extremely high theoretical specific capacity of $3860 \text{ mA}\cdot\text{h}\cdot\text{g}^{-1}$ and the lowest reduction potential (-3.040 V vs. standard hydrogen electrode) [3–5]. However, its practical implementation in liquid LIBs has been hindered by uncontrollable lithium dendrite growth and associated safety concerns. To mitigate the drawbacks of conventional LIBs, solid-state batteries (SSBs) employ more stable solid electrolytes, which not only lower the risks of leakage and ignition but also enable higher energy density [6–9]. SSBs are considered to be the next generation of energy storage devices with advantages such as high

safety and high energy density, and are currently a popular research direction in the field of LIBs. Solid-state electrolytes (SSEs) are a key part of SSBs, for which they need to meet: high ionic conductivity, wide electrochemical stable window, stable interfacial properties and other requirements [10–12]. Inorganic electrolyte, including oxide-, halide-, and sulfide-based materials, outperform polymer electrolytes in ionic conductivity, thermal stability, and mechanical strength [13–15]. Their favorable properties position inorganic SSEs as strong candidates for widespread adoption in SSBs. Current research focuses on further improving their ionic conductivity to rival or even surpass that of liquid electrolytes, enhancing their electrochemical stability with electrodes, and optimizing their mechanical properties. The selection of suitable SSEs is currently the main technical bottleneck in the development of SSBs technology.

Researchers have conducted valuable explorations from both experimental characterization and theoretical simulation perspectives,

* Corresponding authors at: Qingdao Industrial Energy Storage Research Institute, Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, Qingdao 266101, China.

E-mail addresses: zhangshu@qibebt.ac.cn (S. Zhang), cuiql@qibebt.ac.cn (G. Cui).

<https://doi.org/10.1016/j.commt.2026.100051>

Received 26 December 2025; Received in revised form 11 February 2026; Accepted 15 February 2026

Available online 26 February 2026

2950-4635/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

synthesizing and discovering excellent SSEs, thus establishing a solid research foundation for the practical application of SSBs [16,17]. However, experimental characterization demands stringent requirements for laboratory conditions, experimental apparatus, and the professional expertise of researchers, leading to longer experimental cycles and an inability to accelerate the progress of materials research. Although theoretical simulation overcomes the drawbacks of experimental measurement methods that rely on specialized equipment, environmental conditions, and expert experience, leading to significant efficiency improvements compared to experimental approaches, properties that can currently be obtained through computation remains limited [18–20]. Furthermore, such simulations are highly dependent on material microstructure and high-performance computing resources, making the theoretical simulation of tens of thousands of materials still a time- and resource-intensive process. In recent years, with the substantial accumulation of experimental, theoretical, and computational data, as well as the rapid development of efficient and accurate artificial intelligence technologies, materials science research has entered the fourth paradigm of science—data-driven materials science research [21, 22]. Currently, as one of the typical representatives of data-driven artificial intelligence methods, Machine learning (ML) has been widely applied in the field of Energy Chemistry [23–28].

With the increasing complexity of solid electrolyte material systems, traditional experimental and theoretical methods face efficiency bottlenecks in exploring vast chemical spaces. High-Throughput Screening (HTS) has emerged as a key strategy for accelerating the discovery of solid-state electrolytes by enabling systematic and large-scale evaluation of material properties [29–34]. However, conventional HTS relies heavily on extensive experimental work or Density Functional Theory (DFT) calculations, which are time-consuming, costly, and limited in scalability—especially for simultaneous assessment of multi-dimensional properties such as ionic conductivity and electrochemical stability windows. ML, by learning the complex mapping relationships between material structures and properties from existing data, can predict the performance of thousands of candidate materials within seconds, significantly enhancing screening efficiency and enabling rapid evaluation and prioritization of material properties [35, 36]. In recent years, ML has attracted substantial interest in the research of SSEs [37–45]. The application of ML in this field has evolved from preliminary explorations to more sophisticated uses, encompassing material screening and design, structural optimization, property prediction, interface engineering, and energy management. These developments have greatly accelerated the pace of innovation in SSEs research. By leveraging ML models, key properties of new SSE materials, such as ionic conductivity, electrochemical stability, and electrode compatibility, can be predicted from their chemical composition and structural features. The application of ML in explaining the mechanisms and predicting the properties of solid-state electrolytes can be primarily categorized into two directions: accelerating theoretical simulations through machine learning molecular dynamics (MLMD) and directly predicting key performance indicators such as ionic conductivity and electrochemical window using ML models. Molecular dynamics simulation methods primarily include classical molecular dynamics (CMD) simulation and *ab initio* molecular dynamics (AIMD) simulation. CMD typically employs potential functions to describe the interactions between atoms [46]. Classical potential functions offer advantages such as simplicity in form and fast computation, but they fall slightly short in terms of computational accuracy and universality. On the other hand, AIMD utilizes quantum mechanics methods to describe interatomic interactions, providing high computational accuracy and excellent universality [47,48]. However, AIMD suffers from slow computation speeds and significant limitations in the size of systems it can simulate. To address these challenges, MLMD approaches involve solving a certain number of Schrödinger equations to construct a database of material structures, energies, and forces for the studied system [27,49]. ML models are then used to establish quantitative relationships between

material structures and their corresponding energies and forces, thereby constructing ML potential functions. Finally, these ML potential functions are employed to conduct molecular dynamics simulations. The MLMD method holds the promise of maintaining the computational accuracy of AIMD while approaching the computational speed of CMD [39,50–53]. The conventional workflow for predicting solid-state electrolyte properties using ML typically follows a systematic, data-driven pipeline. It begins with the collection and featurization of material data, where key descriptors—such as compositional attributes, structural parameters (e.g., lattice constants, coordination environments), and physicochemical features, which extracted from experimental or computational databases. These features are then used to train ML models (e.g., random forest, gradient boosting, or graph neural networks) to establish quantitative structure-property relationships. The trained model can predict target properties, such as ionic conductivity, electrochemical stability, and interface compatibility, for new materials. To enhance reliability, the process often incorporates validation through first-principles calculations or experimental synthesis, while techniques like cross-validation, uncertainty quantification, and active learning are employed to address data scarcity and improve predictive accuracy.

In contrast to reviews that broadly discuss ML algorithms or their general applications in materials science, this article provides a field-specific perspective focused squarely on inorganic SSEs. We will not systematically introduce the fundamental theories of ML or MLMD, as these topics have been extensively covered elsewhere [41,53–55]. Instead, the primary objective of this review is to systematically organize, analyze, and critique the latest applications of ML techniques in accurate performance prediction and rational screening or design of new SSEs. We will highlight successful case studies, key challenges, and future opportunities. By offering this focused analysis, we aim to provide a practical reference for researchers in the field and promote the deeper integration of the data-driven paradigm in advancing SSEs.

2. Applications of ML in stability of SSEs

Stability is a fundamental prerequisite for the practical application of SSEs. Ensuring thermodynamic stability and electrochemical inertness is crucial for battery safety and long-term cyclability. Traditional methods for evaluating these stability metrics are often time-consuming and computationally intensive, hindering rapid material discovery. By learning complex mappings between material descriptors and stability properties from existing data, ML offers a powerful tool for high-throughput prediction and screening. This section systematically reviews recent advances in ML applications for predicting key stability indicators of SSEs, including thermodynamic stability and electrochemical stability window, highlighting how data-driven approaches accelerate the discovery of stable and reliable SSEs.

2.1. Thermodynamic stability

Any prospective SSEs must exhibit the capability to establish a thermodynamically stable solid phase, which is crucial for practical, large-scale implementation. However, experimentally evaluating thermodynamic stability is often a challenging and time-consuming process. Atomistic modeling offers a means to enhance the efficiency of this task by predicting the propensity for phase formation and assessing stability, thereby providing valuable guidance for subsequent experimental efforts in synthesis and device development. Among the most widely adopted descriptors, the energy above the convex hull (E_{hull}) provides a quantitative metric of synthesizability, representing the distance of a compound's composition from the stable phase boundary in a multi-component system. While DFT calculation can compute E_{hull} accurately, its high computational cost limits broad chemical exploration. By learning from existing computational or experimental datasets, ML models can rapidly predict thermodynamic stability indicators such as E_{hull} and formation energy (E_f) based on compositional or structural

descriptors, dramatically accelerating materials screening.

Ceder et al. leverages a combination of high-throughput first-principles calculations and ML to predict the stability of NASICON materials [56]. Using DFT, they calculated the E_{hull} for 3881 potential NASICON compositions. This metric serves as a direct indicator of synthesizability: SSEs with high E_{hull} values are typically considered thermodynamically unstable and unlikely to be synthesized via conventional methods. They then applied Sure Independence Screening (SIS) and machine-learned ranking (MLR) to identify a simple two-dimensional descriptor that effectively separates stable from unstable compounds. The resulting model, based on features such as Na content, elemental electronegativities, ionic radii, and Madelung energy, yields a linear decision boundary for synthetic accessibility at 1000 K. This data-driven approach provides a quantitative and interpretable stability rule, demonstrating how ML can accelerate the discovery of complex materials beyond empirical or purely phenomenological models.

In addition to E_{hull} , formation energy is also an important criterion for measuring thermodynamic stability [57]. E_f reflects the thermodynamic tendency for a compound to form from its elemental constituents in their standard states. It has been demonstrated by Ceder et al. that existing ML models can predict E_f with considerable accuracy based solely on chemical composition [58]. Jiang et al. developed a feature set based on compositional and structural descriptors, incorporating elemental properties and crystallographic information [38]. This feature set was used to train an XGBoost regression model on a dataset of known materials whose E_f were precomputed using DFT calculations. The model effectively learns the underlying relationships between the input descriptors and the target formation energies. Once trained, it enables the rapid and accurate prediction of E_f for new or hypothetical material compositions, facilitating efficient high-throughput screening of candidate materials without the need for costly DFT calculations for each candidate. Dong et al. successfully applied the ROOST model to predict

the E_f and E_{hull} for anti-perovskite SSEs (Fig. 1) [35]. The model achieved a MAE of 0.08 eV/atom and RMSE of 0.10 eV/atom for E_f , demonstrating reasonable accuracy and consistency. Compounds were selected based on the criteria of $E_f \leq -1$ eV/atom and $0.0 \leq E_{\text{hull}} \leq 0.2$ eV/atom, which confirm their thermodynamic stability and reliability.

2.2. Electrochemical stability

Thermodynamic stability lays the groundwork for both synthesizability and long-term operational integrity of a material. While essential, it represents only a foundational requirement in the practical deployment of SSEs. Beyond thermodynamic stability, SSEs must also remain electrochemically inert to withstand detrimental side reactions with the electrodes, particularly with the highly reducing lithium metal anode and high-voltage cathodes. The electrochemical stability window (ESW) represents the voltage range relative to lithium metal within which an SSE remains stable, resisting decomposition due to lithium consumption or release [59]. Its significance is multifaceted, encompassing safety, performance, and practical viability. Operating a battery cell outside this window induces electrolyte decomposition, risking gas evolution, resistive layer formation, and structural failure. A wide ESW is not just a metric but an essential prerequisite for developing reliable, long-lasting, and high-energy SSBs. The accurate measurement and prediction of ESW in SSEs poses significant challenges, as traditional experimental and computational methods are not only time-consuming but can also suffer from inaccuracies due to complex interfacial effects and approximations in simulations. This bottleneck significantly slows down the discovery of viable SSEs. In this context, ML offers a powerful and efficient alternative. By leveraging large datasets of material properties and known ESW values, trained models can rapidly predict the stability windows of new or hypothetical compounds. These predictions guide the prioritization of

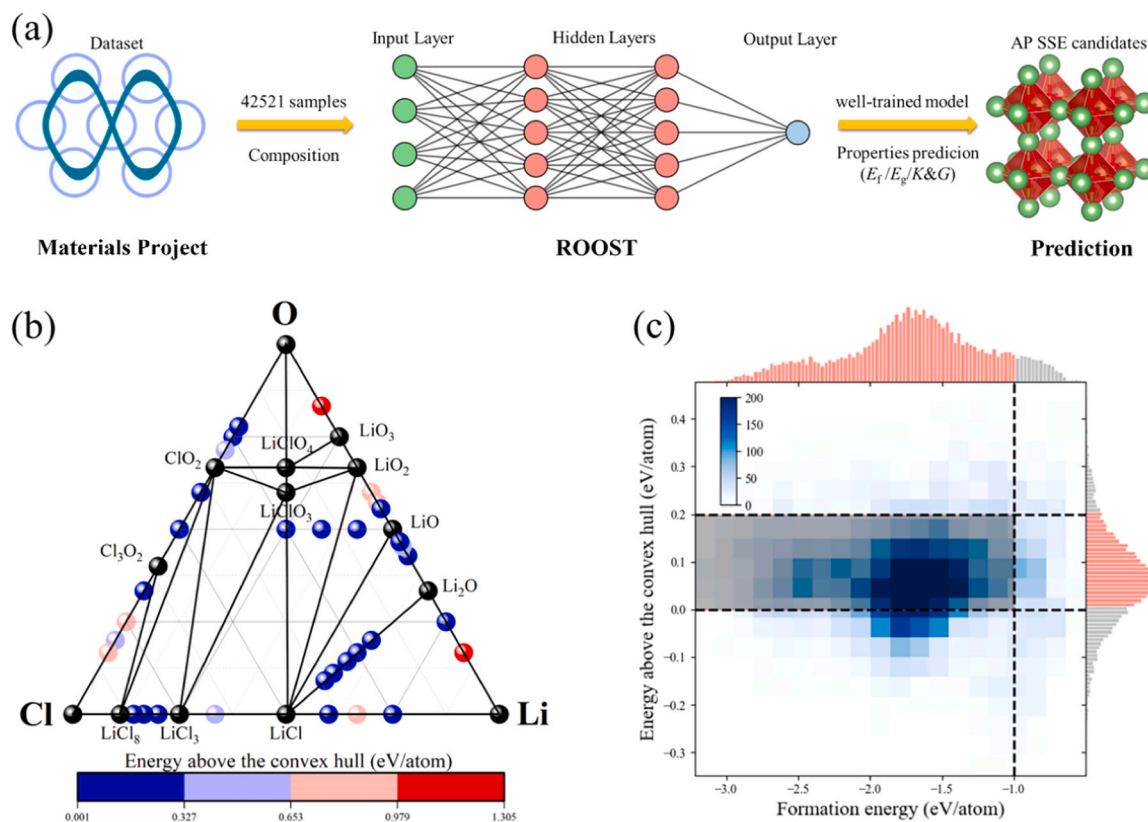


Fig. 1. (a) Workflow of ROOST CML for predicting anti-perovskite SSE candidates. (b) Phase diagram of the Li-O-Cl system depicting the E_{hull} . Black spheres represent stable phases with E_{hull} equal to 0 eV/atom. (c) Heatmap illustrating thermodynamic stability screening. The marginal histograms show the distributions of E_f and E_{hull} , while shaded rectangular region denotes the area that satisfies the screening criteria [35].

promising candidates for synthesis and experimental validation, substantially accelerating the discovery cycle. Moreover, ML models can uncover complex, non-linear relationships between a material's composition, structure, and its electrochemical stability, which are difficult to derive through conventional approaches. Thus, integrating ML into the workflow not only mitigates the challenge of ESW prediction but also transforms it into a strategic tool for the rational and accelerated design of next-generation SSEs.

Yang et al. predicted ESW through a multi-feature regression ML approach using the CatBoost model [45]. After screening 21,686 lithium-containing inorganic compounds from the Materials Project database, the model was trained with descriptors extracted via the Magpie package, augmented by four oxidation-state-based features—including Eng_{max} and Eng_{min} , defined as the product of electronegativity and maximum/minimum oxidation states, respectively. The CatBoost regressor achieved high prediction accuracy, with R^2 values of 0.927 for oxidation potential and 0.908 for reduction potential on the test dataset (Fig. 2a). SHAP analysis identified Eng_{max} , space group number, and electronegativity as the most influential features for oxidation potential, while electronegativity and Eng_{max} significantly affected reduction potential. The ESW was then calculated as the voltage range between the predicted oxidation and reduction potentials, enabling efficient identification of SSEs with desirable electrochemical stability for high-performance SSBs.

The current landscape of ESW datasets suffers from constraints in both scale and chemical variety, which impedes thorough analysis and limits the practical use of data-driven approaches. One major issue is

that existing investigations tend to concentrate on a limited range of materials, usually highlighted for their superior ionic conductivity, thereby neglecting a vast portion of the potential chemical landscape. Additionally, published literature frequently prioritizes reporting positive results while underrepresenting unsuccessful or suboptimal findings, resulting in a dataset that lacks the breadth necessary for training reliable predictive models. Consequently, there is an urgent demand to establish an ESW database to advance the data-driven exploration and design of materials with wide ESW. To address this, Hou et al. established a comprehensive ESW database containing over 16,000 lithium-containing compounds [61]. The ESW values were computed by constructing lithium grand potential phase diagrams (GPPDs) in equilibrium with a lithium reservoir, using density DFT data sourced from the Materials Project database. A composition-based ML framework was developed, employing both classification and regression models. The Roost regression model demonstrated high accuracy, achieving R^2 values of 0.89 and 0.85 for predicting the left and right ESW endpoints, respectively, with mean absolute errors as low as 0.19/0.21 V (Figs. 2c and 2d). The classification model, designed to identify materials with ESW widths ≥ 3 V, achieved accuracy, precision, recall, and F1 scores all exceeding 0.98, along with an area under the receiver operating characteristic (ROC) curve of 0.95. These models were successfully applied to screen the OQMD database, identifying 74 promising candidate materials, of which seven exhibited both wide ESWs and high predicted ionic conductivity, highlighting the efficacy of this data-driven approach for accelerating the discovery of electrochemically stable SSEs.

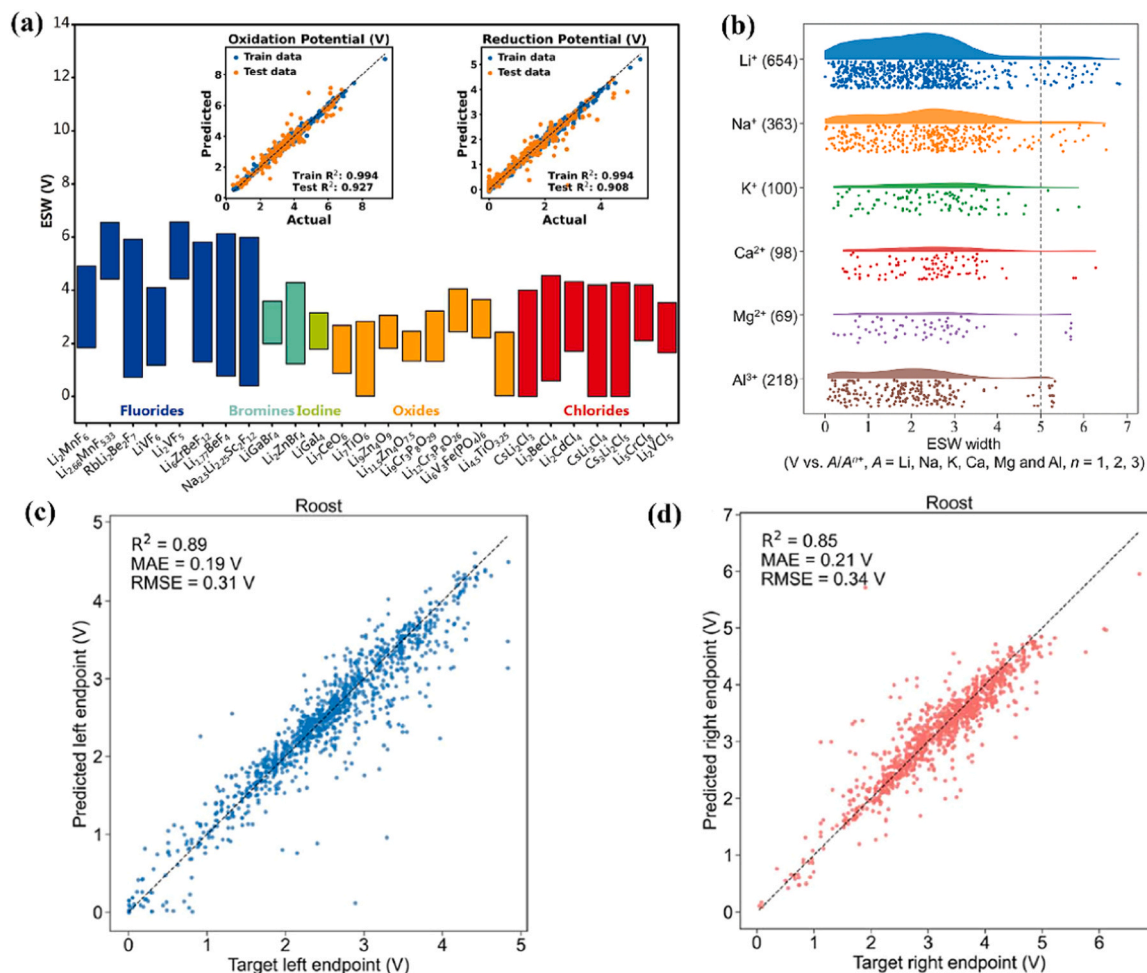


Fig. 2. (a) Active learning model predicted ESW of SSEs [45]. (b) Distribution of ESW widths for inorganic compounds of each migrating species (Li^+ , Na^+ , K^+ , Ca^{2+} , Mg^{2+} , and Al^{3+}) [60]. Parity plots of the Roost regression model for the ESW (c) left endpoint and (d) right endpoint [61].

Shi et al. present a database for more than 1 500 solid state inorganic compounds, including information on the ESW, phase diagrams and reactive decomposition energies of structures (Fig. 2b) [60]. The ESW is predicted using the Direct and Indirect Decomposition Analysis Methods (DDAM and IDAM). DDAM relies solely on thermodynamics and may underestimate ESW by ignoring kinetics, while IDAM accounts for kinetics but may overestimate ESW by neglecting thermodynamics. To overcome these limitations, they developed an integrated approach that dynamically selects decomposition pathways based on the electronic conductivity of products, achieving higher prediction accuracy for ESW. The database is available on the Electrochemical Energy Storage Materials Design Platform (EESMDP), and new entries will be added continuously.

3. Application of ML in charge transport properties of SSEs

After establishing thermodynamic and electrochemical stability, the development of SSEs must address their core charge transport

characteristics. These properties define a fundamental dual requirement: minimal electronic conduction to prevent self-discharge and short circuits, alongside efficient ionic transport to ensure sufficient power density and battery efficiency. Optimizing this balance is critical for replacing liquid electrolytes and unlocking the full potential of SSEs. This section reviews how ML aids in understanding and predicting both electronic and ionic transport phenomena, covering bandgap prediction as a key electronic insulator metric, direct ionic conductivity estimation, migration barrier analysis, and the emerging role of lattice dynamics in ion conduction.

3.1. Electronic transport properties

SSEs must be effective electrical insulator to prevent short circuits, self-discharge, and parasitic reactions at the electrode-electrolyte interface. This necessitates an evaluation of its electronic transport properties. A key determinant of electronic conductivity is the bandgap (E_g), which corresponds to the energy separation between the

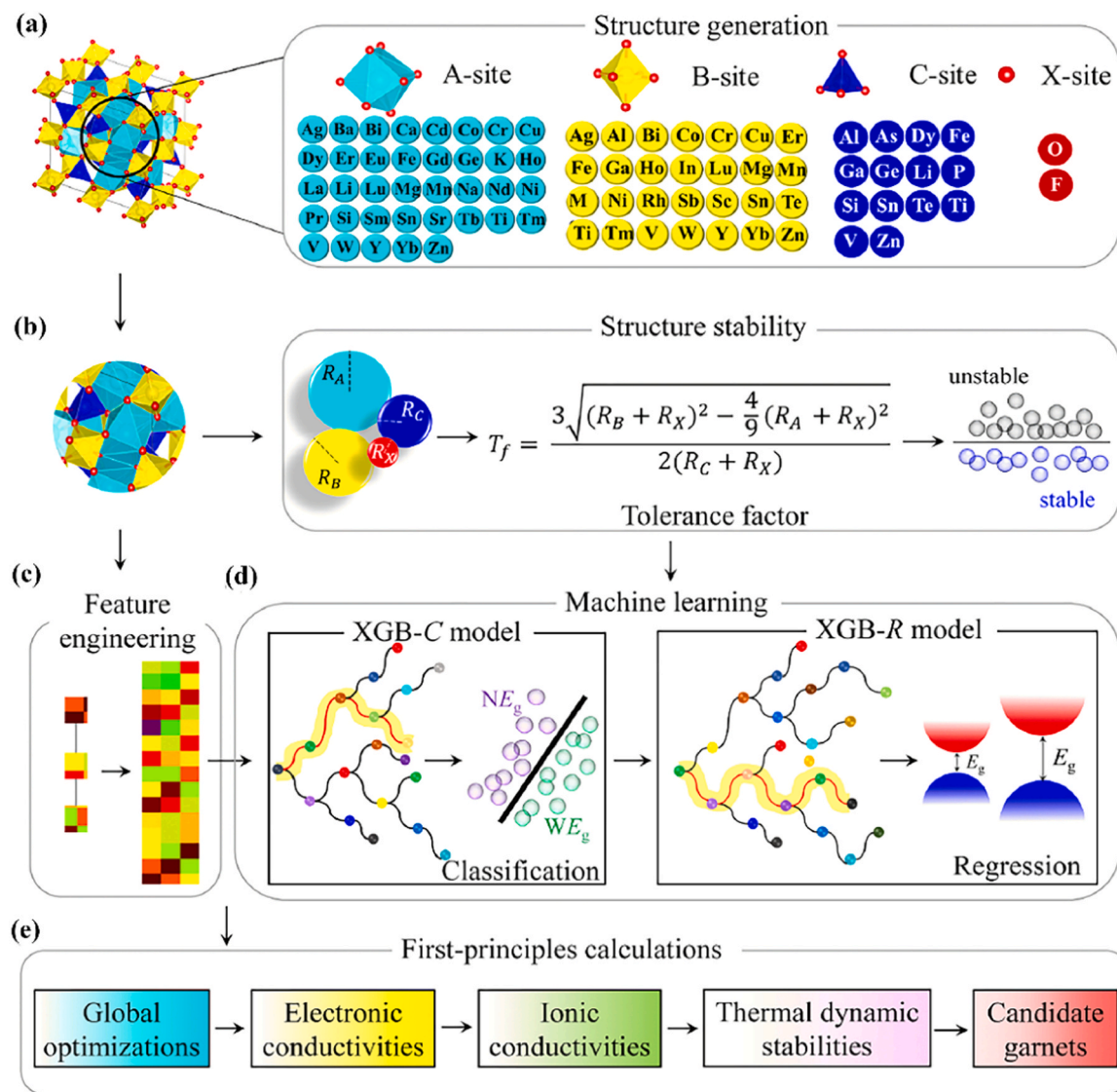


Fig. 3. Overview of the framework to discover garnet SSEs. (a) Left panel, crystal structure of garnet materials with four element sites and space group of Ia3d. Right panel, suitable element selection based on 286 known garnets for structure generation, where 37 elements are selected for the dodecahedron A-site, 28 elements for the octahedral B-site, 14 elements for the tetrahedron C-site, and two elements for the X-site. (b) Preliminary judgment of structural stability by tolerance factor, according to the ionic radius of four sites. (c) Feature engineering based on elemental properties for the ML model. (d) A narrower/wider band gap (NE_g/WE_g) classification model (XGB-C) and a band gap regression model (XGB-R) built for electronic conductivity prediction. (e) Further validation and screening for candidate garnets by first-principles calculations [63].

conduction band minimum (CBM) and the valence band maximum (VBM). Generally, a wide bandgap corresponds to low electronic conductivity, which is a critical requirement for SSEs to effectively block electron transport while allowing ionic conduction. In the high-throughput screening workflow for SSEs, possessing a wide bandgap often serves as the first step in the selection process. High-throughput screening of bandgaps typically relies on data from established computational databases such as the Materials Project. For materials not yet included in existing databases, experimental measurement of bandgaps is time-consuming and labor-intensive. Although databases of DFT calculations are valuable for predicting electronic band gaps, their accuracy is fundamentally constrained by the exchange–correlation functionals employed. A common issue is the significant discrepancy between experimental band gaps and those predicted by the widely used GGA/PBE method [62], which serves as the standard for high-throughput calculations. While the HSE functional performs particularly well for solids, its computational cost is one to two orders of magnitude higher than that of PBE [33]. DFT-based bandgap prediction faces a trade-off between accuracy and computational cost. Therefore, ML methods demonstrate distinct advantages in this context: by training regression or classification models based on compositional or structural features, rapid estimation of bandgaps for novel materials can be achieved while maintaining reasonable predictive accuracy. This approach significantly enhances the efficiency and expands the scope of high-throughput screening.

To obtain large band gap garnets quickly from 7067 structures at the lowest cost, Li et al. established a classification model to classify the narrower band gap (NE_g , $E_g < 0.5$ eV) and the wider band gap (WE_g , $E_g > 0.5$ eV) (Fig. 3) [63]. Then, for the WE_g (179 garnets), they further established a regression model (XGBR) for accurately predicting E_g to quickly determine the large band gap garnets for electrolytes. There is only a minor discrepancy between the E_g values predicted by the XGB-R model and those calculated by DFT for most structures, demonstrating the model's accuracy. The ML model detected 210 novel garnets with low electronic conductivity. Subsequent refinement identified 12 garnets that demonstrated high ionic conductivity, along with thermal stability confirmed by *ab initio* molecular dynamics (AIMD) simulations. By accelerating the discovery process, this methodology reduced the typical screening timeline by an estimated 95 years, yielding substantial cost reductions and accelerating progress in SSBs technology.

3.2. Ionic conductivity

SSEs must exhibit excellent ion transport properties to ensure proper battery operation, with ionic conductivity serving as a key indicator for evaluating their suitability as SSEs. Ionic transport properties represent the ability of lithium ions to move through the solid material, a process which dictates the battery's internal resistance. A solid electrolyte with low ionic conductivity creates a significant kinetic barrier, leading to poor battery performance, inefficient energy use, and excessive heat generation. Therefore, achieving high ionic conductivity is not merely an optimization goal but a foundational requirement for replacing liquid electrolytes and realizing the full potential of SSBs in terms of safety, energy density, and lifespan. The prediction of ionic conductivity, however, presents a major challenge. Traditional methods rely on experimental trial and error or complex atomistic simulations, both of which are prohibitively slow and expensive for exploring the immense landscape of possible inorganic compounds. This is precisely where ML becomes indispensable. Trained on existing data that correlates material composition and structure with ionic transport, ML models can evaluate thousands of hypothetical materials in seconds. They identify promising candidates with predicted high conductivity by recognizing underlying patterns that are difficult for humans to discern. This data driven approach dramatically accelerates the discovery pipeline, shifting the research paradigm from slow, sequential testing to intelligent, high throughput screening. ML approach for predicting ionic conductivity

involves extracting descriptors from the crystal structure of SSEs and establishing a quantitative relationship between the structural features and intrinsic ionic conductivity. Furthermore, interpretability analysis through feature engineering provides rational strategies for designing and developing SSEs with high ionic conductivity. By focusing experimental efforts only on the most viable candidates identified by ML, the development of next generation SSEs is greatly accelerated, saving significant time and resources.

In the early stages, the limited availability of databases meant that training samples were primarily sourced from experimental data and were relatively scarce in quantity. Reed et al. compiled a training set of 40 crystal structures and reported experimentally measured ionic conductivity values available in the literature [64]. Based on physical intuition and previous proposals and reports in the literature, they computed 20 features that characterize the local atomic arrangements and chemistry of the crystals from atomistic structures. Despite the small training set, the logistic regression ML model demonstrated predictive capability significantly better than random guessing through rigorous feature selection and validation. The practical validation of this approach was demonstrated when a high-throughput screening of 12 000 compounds from Materials Project database successfully identified 21 promising fast-conductor candidates, proving the feasibility of extracting universal patterns from complex crystal structures. This early success demonstrated that even with a limited dataset and a relatively simple ML model, when combined with physically meaningful features, it was possible to effectively navigate the vast chemical space. This achievement laid a foundational cornerstone for the subsequent application of ML in predicting SSEs conductivity.

Due to the limited availability of experimental data on the ionic conductivity of SSEs and the strong dependence of model predictions on the types of descriptors selected, Zhang et al. developed an unsupervised ML model to address this issue, enabling the screening and design of SSEs with high ionic conductivity (Fig. 4) [65]. They proposed an innovative unsupervised ML approach to address the core bottleneck of scarce experimental data in the discovery of solid-state lithium-ion conductors. Instead of directly predicting the difficult-to-accurately-obtain ionic conductivity values, the method extracts the geometric features of a material's anionic framework through a novel structural representation (modified X-ray diffraction, mXRD) and performs cluster analysis on thousands of candidate compounds based on this information. The study found that despite the diverse compositions and specific structures of known high-performance conductors, they automatically converge into a few specific groups during unsupervised clustering, revealing a common characteristic of "moderate distortion" in their anionic sublattices. Based on this pattern, candidate materials were screened from the clustered groups and validated via AIMD, successfully identifying 16 new high-performance conductors, three of which have predicted conductivities exceeding 10^{-2} S/cm. This work demonstrates that unsupervised learning can effectively utilize limited structural data, offering a powerful and versatile new paradigm for novel material discovery under conditions of data scarcity.

Hargreaves et al. constructed an expert-curated database of 820 experimentally measured SSEs conductivities, including 403 unique compositions at room temperature [66]. They developed a transfer learning-enhanced CrabNet model that effectively predicted whether a material exhibits high ionic conductivity based solely on its composition, achieving a Matthews Correlation Coefficient of 0.38 under leave-one-cluster-out cross-validation. This practical classification tool aids experimentalists in prioritizing candidate materials for further investigation, while underscoring the critical need for standardized data reporting to advance data-driven discovery in SSEs research.

Shi et al. developed a tool named CAVD (Crystal structure Analysis by Voronoi Decomposition), which employed an improved radical Voronoi decomposition method to characterize the void space in crystals by constructing an interstitial network for ionic transport analysis [67,

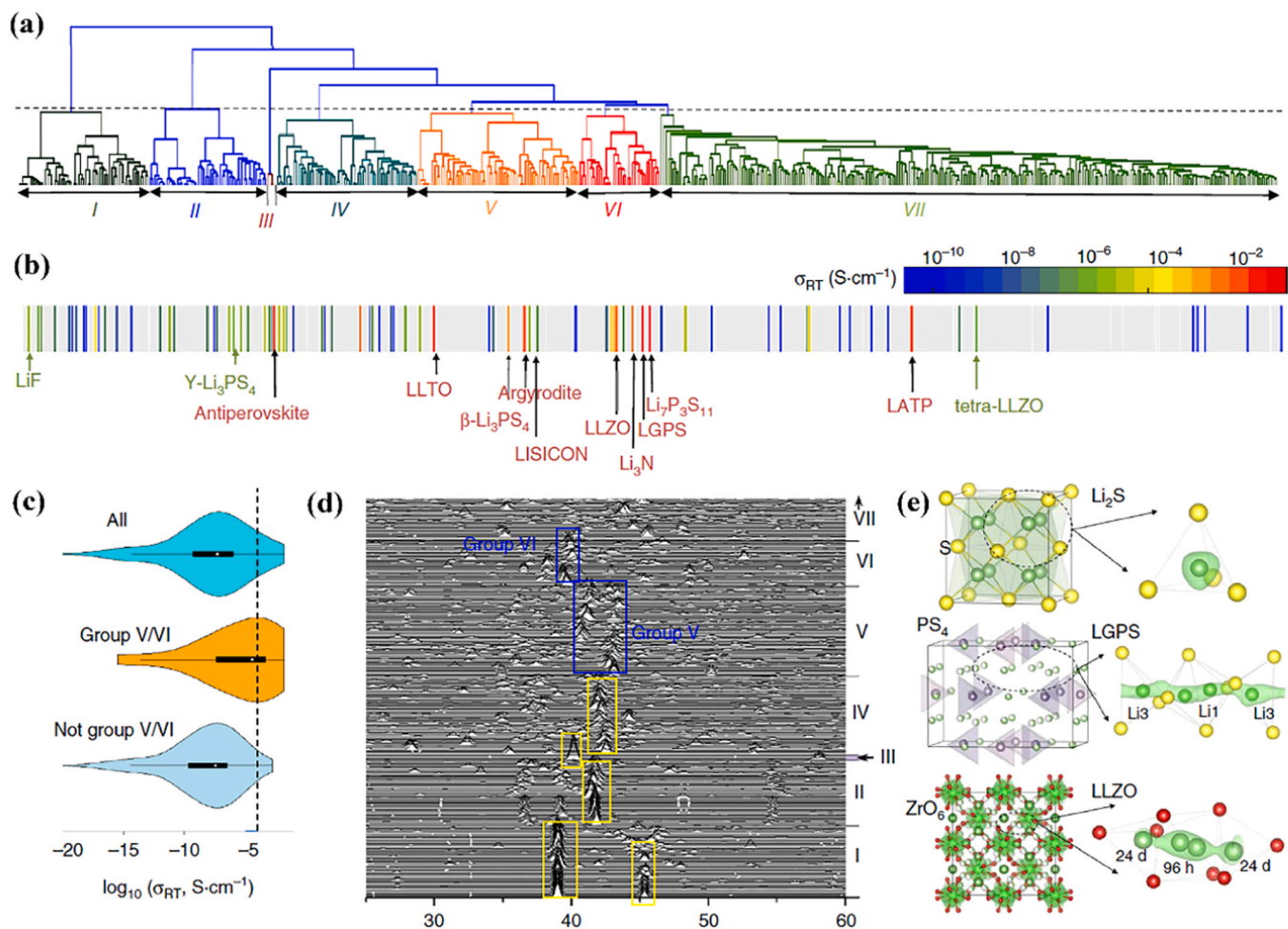


Fig. 4. Unsupervised clustering of all Li-containing compounds. (a) Bottom-up tree diagram (dendrogram) generated using the agglomerative hierarchical clustering method. The dashed line shows the position where all compounds are partitioned into seven groups, marked as I–VII from left to right and distinguished by different colors. (b) Mapping the dendrogram to the conductivity reveals the grouping of known solid-state Li-ion conductors in group V and VI. The color bar shows the scale of σ_{RT} . The gray color indicates the conductivity has not been measured for the corresponding compound. (c) Violin plots of σ_{RT} data grouped in the grouping. The outer shells of the violins bound all data, narrow horizontal lines bound 95% of the data, thick horizontal lines bound 50% of the data, and white dots represent medians. The dashed line shows the position of $\sigma_{RT} = 10^{-4}$ S cm $^{-1}$. (d) mXRD of all materials in group I–VI and a part of group VII. The colored boxes mark the positions of main characteristic peaks for each group. (e) Crystal structures (left) and (right) Li sites (green sphere) determined by local anion (yellow/red sphere) configuration, corresponding to isosurfaces (green) of Li probability density from AIMD simulations. Li_2S (top) with highly symmetric anion lattice and ordered Li sublattice versus LGPS (middle) and LLZO (bottom) SSLCs with distorted anion lattices and disordered Li sublattices [65].

[68]. By analyzing 6955 Li-, Na-, Mg-, and Al-containing ionic compounds extracted from the Inorganic Crystal Structure Database, CAVD achieves a high recovery rate of up to 99% for the lattice sites of mobile ions and can compute ionic transport descriptors such as transport channels, network dimensionality, and critical ionic radii. The study further validated the effectiveness of CAVD for various typical ionic conductor materials, demonstrating its utility for high-throughput screening, structural classification and ML. They developed a high-throughput SSEs screening platform named SPSE, which integrates a materials database with hierarchical ion-transport calculation algorithms to accelerate the discovery and design of high-performance SSEs [69]. SPSE combines geometric analysis (CAVD) and bond valence site energy (BVSE) methods to rapidly simulate ion migration pathways and energy profiles, thereby automatically generating approximate minimum energy paths. This provides preprocessed input for first-principles nudged elastic band (FP-NEB) calculations without manual intervention. The platform contains approximately 91,763 crystal structure data entries and implements an automated workflow from preliminary to fine-tuned screening through a hierarchical calculation process. It efficiently and automatically identifies ion transport pathways and migration energy barriers, offering a powerful high-throughput computational

tool for SSBs materials research.

3.3. Migration barrier

From the Nernst-Einstein equation and the Arrhenius equation, established that the microscopic ion migration barrier (E_a) is one of the most critical physical quantities determining ionic conductivity, with the two exhibiting an exponential inverse relationship [18]. Reducing E_a is a fundamental strategy for enhancing conductivity, and the accurate prediction of a material's migration barrier effectively forecasts its ionic transport properties. Shi et al. proposed a ML framework based on hierarchically encoding crystal structure-based (HECS) descriptors for predicting E_a of SSEs (Fig. 5a) [70]. Using cubic Li-argyrodites as an example, the research constructed 32 HECS descriptors covering five categories: composition, structure, conduction pathways, ion distribution, and special ions. A partial least squares (PLS) model for E_a prediction was established, achieving coefficients of determination (R^2) of 0.887 and 0.820 on the training and test sets,

respectively, with a prediction error of only 0.02 eV, demonstrating the effectiveness of the HECS descriptors. To elucidate the chemical and structural determinants of ion transport in crystalline lattices, Kim et al.

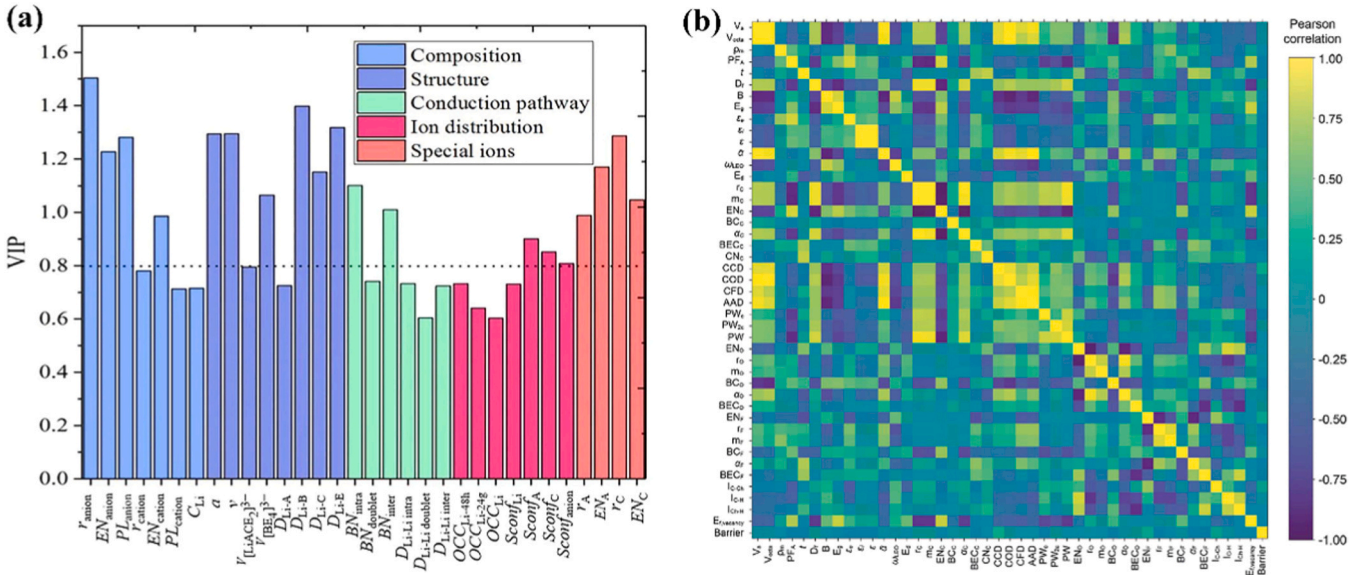


Fig. 5. (a) VIP scores of descriptors (explanatory variables) in the partial least squares model. Five different color areas correspond to five categories. The dotted line at 0.8 represents the default cut-off line of the JMP software [70]. (b) Pearson correlation analysis for descriptors and vacancy migration barriers [43].

employed ML to systematically identify and quantify the key factors governing cation migration barriers in anti-perovskite SSEs (Fig. 5b) [43]. This study analyzed over 600 DFT-calculated energy barriers across 36 alkali metal chalcogenide anti-perovskites, considering both vacancy and interstitial migration mechanisms. It revealed that for vacancy migration, the dominant factors are the structural features of hopping distance and migration channel width, which collectively account for approximately 70% of the feature importance. It also reaffirmed the role of anion polarizability and notably uncovers the defect formation energy as a significant, previously overlooked descriptor for interstitial mobility. The optimized predictive models, built using only a handful of non-redundant features, demonstrated that shorter migration paths and wider channels lower energy barriers, providing clear, interpretable design rules.

3.4. Phonons dynamics

The classical ionic conduction model describes diffusion as a series of discrete ionic hops across a static potential energy surface spanning the crystal lattice. It is generally held that a smooth energy landscape

facilitates ionic transport [73,74]. However, the exceptionally high ionic conductivity observed in certain materials cannot be fully accounted for within this theoretical framework [75]. Ling revealed a novel mechanism of phonon-promoted superionic conduction in fluorite-structured compounds, where geometric frustration arises from the mismatch between bonding distances in tetrahedral and octahedral interstices within a face-centered cubic lattice (Fig. 6a) [71]. This frustration drives mobile ions to interact dynamically with lattice phonons, leading to a coupled phonon-ion interaction that continuously modulates the potential energy landscape and facilitates the formation of metastable defects. As a result, the ionic conductivity exhibits pronounced non-Arrhenius behavior, which is quantitatively captured by a phenomenological model incorporating a material-dependent exponent m that encodes the strength of phonon-ion coupling. The work demonstrated that such phonon-enhanced defect formation enables superionic conduction even in compact, symmetrically ordered crystals, expanding the search for high-performance SSEs beyond traditionally considered structural motifs. The conventional thermodynamic equilibrium framework frequently applied to ion migration models in solid-state systems fails to accurately describe the actual Li^+ diffusion

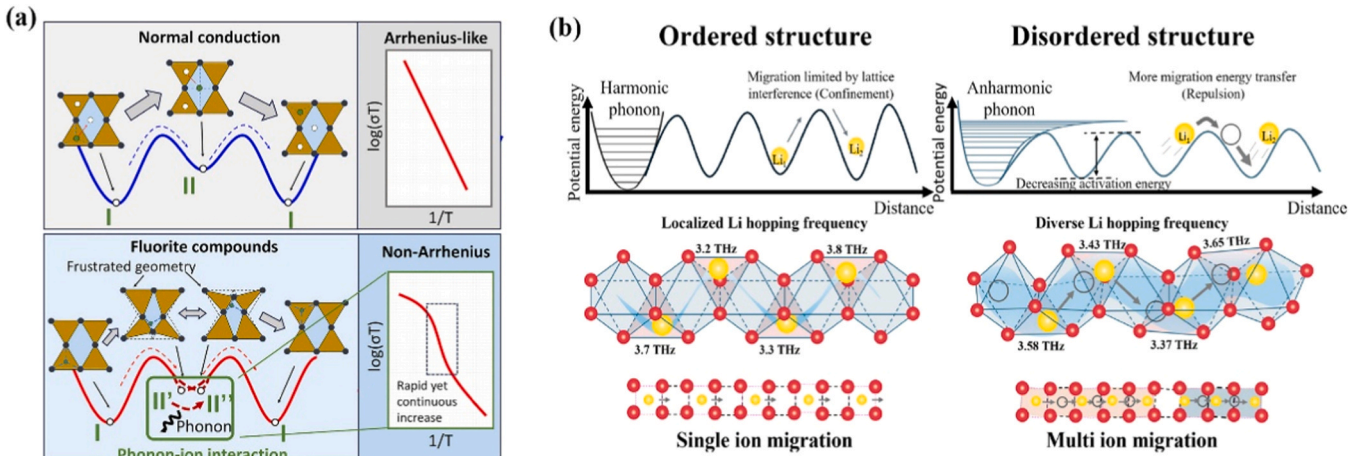


Fig. 6. (a) Phonon-ion interactions are responsible for promoting the formation of anomalous defects, thereby facilitating ion migration [71]. (b) Li ion migration mechanism driven by anharmonic phonon enhancement [72].

behavior, especially in low-crystallinity superionic conductors that demonstrate unexpectedly high ionic conductivity. These materials defy traditional theoretical descriptions, demanding an alternative viewpoint grounded in lattice dynamics that directly incorporates nonequilibrium phonon interactions and transient structural adaptations. Kim et al. uncovered a phonon-governed Li ion migration mechanism in garnet-structured superionic conductors by comparing Ta-doped $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Ta}_{0.4}\text{O}_{12}$ to its undoped analogue, $\text{Li}_{6.24}\text{La}_3\text{Zr}_{2}\text{Al}_{0.24}\text{O}_{11.98}$ [72]. This work revealed that anharmonic lattice vibrations can act as a key driver for achieving ultrafast Li^+ transport in SSEs (Fig. 6b). This conceptual shift established a foundational connection between lattice thermodynamics and superionic conductivity, offering both a theoretical and experimental basis for the rational design of advanced SSEs with high ionic conduction. Growing research indicates that phonons play a decisive role in ionic conduction [75,76]. Their spectral, mode, frequency, amplitude, lifetime, and coupling characteristics are emerging as critical descriptors for evaluating and predicting high ionic conductivity in SSEs.

Jaafresh et al. introduced an efficient ML framework for screening SSEs (Fig. 7) [77]. Using the total phonon band center as a physical proxy for ionic conductivity, the model is trained on phonon density of states data from approximately 1272 compounds in the Materials Project database. The feature set is constructed using the Magpie platform, capturing both compositional and crystal structure information. The extra random trees (XT) algorithm delivers the best performance with an R^2 of 0.964 in cross-validation. The model is validated via DFT calculations on several unseen compounds and experimentally using electrochemical impedance spectroscopy, confirming accurate predictions for known ionic conductors like $\text{Li}_6\text{PS}_5\text{Cl}$ and identifying promising candidates such as LiBiO_2 with a low migration barrier of 0.27 eV. The framework enables high-throughput screening of around 17,114 Li-containing materials and 230,136 virtual structures derived from known electrolyte prototypes, ultimately identifying over 136,041

candidates with a total phonon band center below 40 meV. Kim et al. developed ML models for predicting ionic conductivity in SSEs by integrating dynamic lattice properties with traditional structural and electronic descriptors [37]. They curated 14 phonon-based features from first-principles calculations alongside 16 static descriptors, and applied logistic regression for classification and random forest for regression. The logistic model achieved 93% accuracy in distinguishing superionic conductors, while the regression model yielded a root-mean-square error of 1.179 S/cm for $\log(\sigma)$ and an R^2 of 0.710. Phonon-related features, such as the ratio of mobile-ion phonon DOS below 30 meV and vibrational entropy, were identified as critical predictors. The authors further screened 264 Li-containing materials, proposing 11 promising superionic candidates, and demonstrated that incorporating lattice-dynamics descriptors significantly enhances the predictive power of ML models for SSEs design.

4. Application of ML in mechanical properties of SSEs

The mechanical properties of SSEs represent a cornerstone for advancing safe, durable, and high-performance SSBs. Their significance extends far beyond simple structural integrity, governing critical electrochemical, interfacial, and long-term operational behaviors. This importance manifests primarily in three interconnected domains: First, the modulus of an SSE dictates its ability to mechanically suppress lithium dendrite growth, which is a major failure mechanism in batteries employing lithium metal anodes. According to the Monroe-Newman criterion, a shear modulus exceeding twice that of lithium metal (approximately >8.5 GPa) is theoretically required to prevent dendrite penetration through the electrolyte [78]. A high bulk modulus further reinforces this resistance by reducing the likelihood of local deformation and pore collapse under lithium plating stress. Without this mechanical barrier, metallic filaments can traverse the electrolyte, leading to internal short circuits, thermal runaway, and catastrophic cell failure.

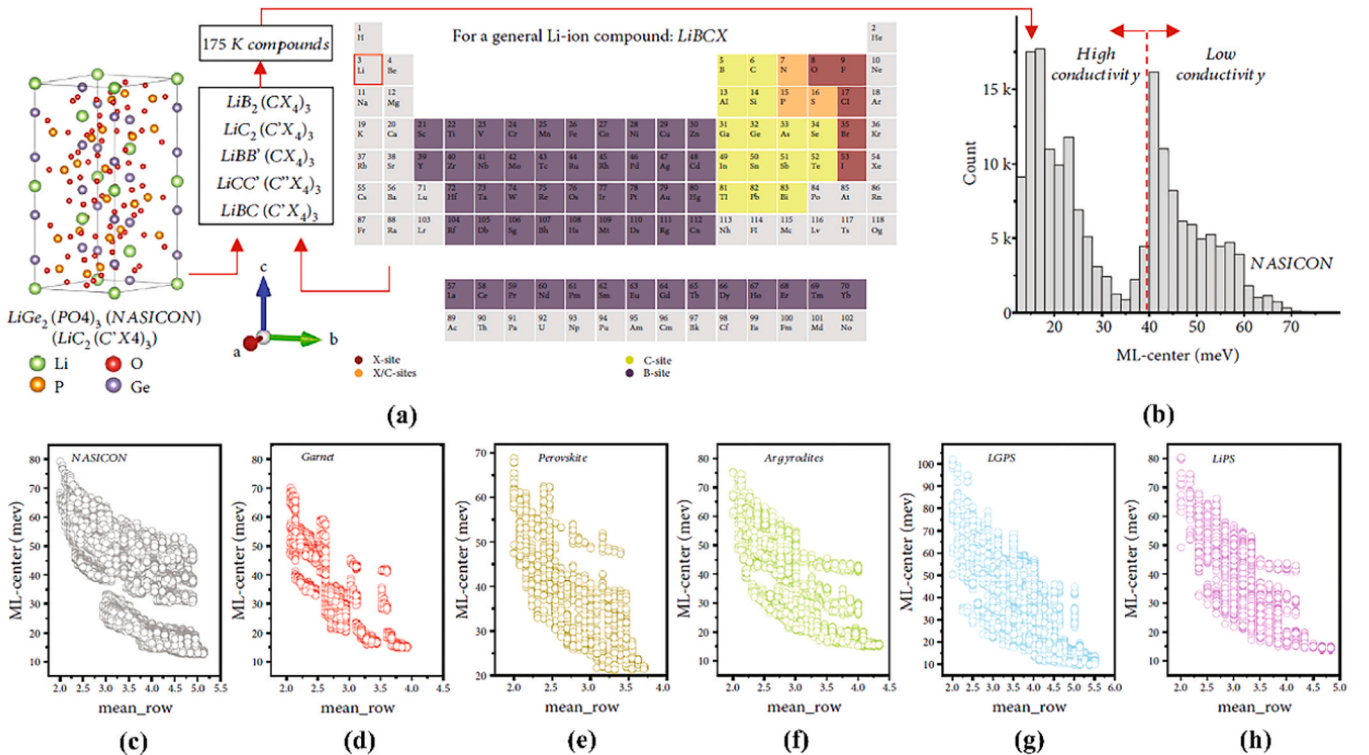


Fig. 7. (a) The structure of $\text{LiGe}_2(\text{PO}_4)_3$ used as a prototype to generate new virtual structures with nonrelaxed structures, for which the total phonon band center is to be predicted using the XT-model built in this work, and the related periodic table which is highlighted to show the chemical substitution in this structure. (b) Histogram of the total phonon band center, as predicted by the XT-model for the nonrelaxed structures constructed based on the NASICON-structured: $\text{LiGe}_2(\text{PO}_4)_3$ (R3c). (c–h) The predicted values of the total phonon band center of the nonrelaxed structures constructed based on the various prototypes [77].

Second, mechanical properties are paramount for ensuring stable and low-resistance interfaces throughout battery cycling. Electrodes undergo significant volumetric changes during lithiation and delithiation. A favorable bulk-to-shear modulus ratio ($K/G > 1.75$) indicates a degree of ductility or "mechanical forgivingness" [79,80]. This ductility allows the SSE to accommodate electrode expansion and contraction without brittle fracture or permanent detachment, thereby maintaining intimate interfacial contact. The loss of such contact creates high local impedance, accelerates uneven current distribution, and ultimately leads to rapid capacity fade. Furthermore, good interfacial adhesion under operational stack pressure is essential for facilitating efficient Li-ion transport across the solid-solid junction. Third, the overall mechanical integrity of the SSE governs its processability and long-term durability. During cell fabrication, the electrolyte must withstand handling stresses without cracking. Throughout extended cycling, external stack pressure and internal stresses from repeated volume changes and thermal fluctuations can initiate and propagate micro-cracks within a brittle electrolyte. These cracks not only compromise structural cohesion but also create new surfaces and block ionic conduction pathways, drastically increasing cell resistance. Therefore, a balanced mechanical profile encompassing sufficient stiffness to resist dendrites, adequate toughness to resist crack propagation, and appropriate hardness to withstand processing is essential. While high ionic conductivity and broad electrochemical stability windows are the most highlighted properties of SSEs, their mechanical performance is an equally critical, non-negotiable design parameter. Optimizing the mechanical profile, achieving a delicate balance between stiffness, ductility, and toughness, is a fundamental prerequisite for realizing the full potential of ASSBs. It enables the safe use of high-capacity lithium metal anodes, ensures stable interfacial kinetics, and guarantees long-term cyclability, collectively paving the way for next-generation batteries with superior energy density, enhanced safety, and extended operational lifespan.

Given the critical role of mechanical properties, the search for optimal SSEs necessitates efficient screening across vast chemical spaces. This challenge has spurred the development of ML approaches to

accelerate the discovery of mechanically superior SSEs. Min et al. developed a data-driven ML approach to identify SSEs with superior mechanical properties to suppress dendrite formation and ensure interfacial stability [81]. From an initial pool of 17,619 candidate materials from the Materials Project database, the authors first constructed a training database using elasticity data (14,238 structures) and engineered 277 descriptors comprising physicochemical, structural, and basic material properties. A LightGBM regression model was trained to predict shear modulus (G_{VRH}) and bulk modulus (K_{VRH}), achieving R^2 scores of 0.852 and 0.906, respectively, with physical features such as volume and density being the most influential (Fig. 8). The model was then applied to predict the mechanical properties of 2842 screened Li-containing SSE candidates, and the predictions were validated with DFT calculations. To improve prediction accuracy, an active learning strategy was implemented by iteratively adding high-uncertainty candidates to the training set. This approach significantly enhanced model performance: adding only 1600 and 1400 materials for G_{VRH} and K_{VRH} , respectively, increased R^2 from ~ 0.63 to over 0.80, outperforming random data addition. The study demonstrated that oxide-based structures generally exhibit higher moduli, while NASICON- and garnet-type SSEs showed relatively lower mechanical strength. The established workflow provided an efficient, uncertainty-aware platform for accelerating the discovery of mechanically robust SSEs.

Building on this methodology, Min et al. employed a similar ML-driven workflow to specifically identify superior garnet-type SSEs with superior mechanical properties [82]. The process begins by training a LightGBM model on a dataset of 17,051 materials using 271 chemical and structural descriptors to predict K_{VRH} and G_{VRH} . Starting from a base $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) structure, 5329 candidate materials are generated via La- and Zr-site substitution with 73 elements. After initial screening based on E_f and E_g via DFT, the LGBM model predicts the elastic properties of the remaining candidates. To enhance prediction accuracy for the underrepresented tetragonal garnet phase, an active learning loop is implemented: materials with high prediction uncertainty are selected, their true elastic properties are computed with DFT, and the model is

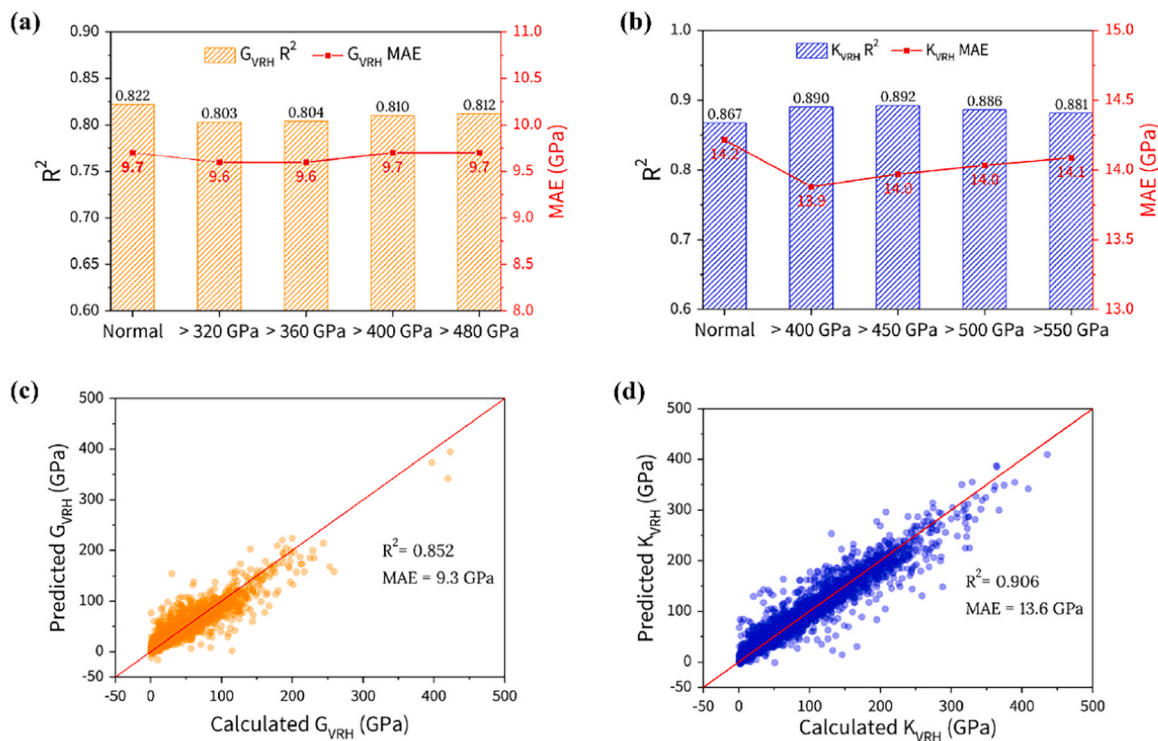


Fig. 8. Prediction accuracy change depending on mechanical property range in the training set for predicting (a) G_{VRH} and (b) K_{VRH} and calculated vs. predicted plot for (c) G_{VRH} and (d) K_{VRH} of the best model among 10 predictions [81].

iteratively retrained. Candidates are then filtered using mechanical criteria: $G_{\text{VRH}} > 8.5$ GPa (dendrite suppression) and $K_{\text{VRH}}/G_{\text{VRH}} > 1.75$ (interfacial stability). This yielded 758 mechanically robust garnet-type materials. Subsequent screening with a separate ionic conductivity classifier identified 29 potential superionic conductors, of which 10 are validated via AIMD simulations. Key results include the discovery of high-performance electrolytes such as $\text{Li}_7\text{Ti}_3\text{Pm}_2\text{O}_{12}$ ($G_{\text{VRH}} = 32.98$ GPa, ionic conductivity = 11.21 mS/cm), demonstrating the effectiveness of this ML-enhanced approach in rapidly discovering mechanically superior SSEs.

Complementing these descriptor-based models, Viswanathan and colleagues employed a structure-based ML approach to screen and predict the mechanical properties of SSEs for dendrite suppression [83]. A Crystal Graph Convolutional Neural Network (CGCNN) was trained on structural data from the Materials Project database to predict G_{VRH} and K_{VRH} , utilizing only crystal structure information without costly first-principles calculations. For predicting the full elastic tensor (C_{11} , C_{12} , C_{44}), particularly for cubic crystal systems, the authors used gradient boosting regression and kernel ridge regression, selected based on their ability to handle varying levels of noise in the training data. These ML-predicted mechanical properties were then fed into a theoretical stability parameter (χ) model to evaluate the dendrite initiation propensity at Li metal-solid electrolyte interfaces, enabling high-throughput computational screening of over 12,000 compounds and identifying promising candidates with desirable mechanical softness and high anisotropy.

5. Application of ML in electrode-electrolyte interface

As an integrated system, battery performance depends not only on the intrinsic properties of each component but also critically on their mutual interactions. Beyond the inherent characteristics of SSEs, the formation and evolution of the interfacial region of electrode-electrolyte play a vital role in ensuring cycling stability and extending battery lifespan. While substantial progress has been achieved in enhancing the bulk ionic conductivity of SSEs, the overall conductivity in ASSBs is frequently compromised by significant interfacial impedance. This elevated impedance raises the overpotential, resulting in capacity decay and reduced energy density. The origins of such interfacial resistance are multifaceted: (1) Incomplete physical contact between electrode and solid electrolyte. Despite applying high pressure or sintering during cell assembly, ideal close-packing remains difficult to achieve. Voids emerge not only during fabrication but also during cycling due to mechanical stresses induced by electrode volume changes. (2) Space-charge layer formation at the cathode-electrolyte interface, arising from differences in lithium chemical potential. Lithium ion accumulation on the cathode side and depletion on the electrolyte side create localized carrier imbalances that impede Li^+ transport across the interface. (3) Chemical reactions between electrode and electrolyte materials driven by chemical potential differences, leading to the formation of solid electrolyte interphase (SEI) and cathode electrolyte interphase (CEI) layers with poor ionic conductivity. (4) Electrochemical decomposition of solid electrolytes due to their limited electrochemical stability windows. These degradation processes further contribute to resistive interphases and exacerbate interfacial impedance. (5) Lithium dendrite growth at interface of electrolyte and lithium metal anode. These needle-like lithium deposits can penetrate through the electrolyte, potentially reaching the cathode and triggering short circuits.

Addressing these complex interfacial challenges requires advanced tools that can capture atomic-scale phenomena over experimentally relevant length and time scales. ML-enhanced interatomic potentials have emerged as a powerful approach to bridge this gap. Ong et al. employed a ML-enhanced computational strategy to predict the behavior of the solid electrolyte-cathode interface in all-solid-state Li-S batteries [84]. A machine-learned interatomic potential is developed for the $\beta\text{-Li}_3\text{PS}_4/\text{S}_8$ interface through an active learning workflow. The

process began with passive training using DFT-relaxed structures and AIMD trajectories of bulk phases and strained slabs, followed by active learning where MD simulations on various interface models are performed. Structures with high extrapolation grades were iteratively selected, validated with DFT, and added to the training set until the MTP achieves high accuracy (energy error < 5 meV/atom, force error < 0.2 eV/Å) and 100% MD reliability. Using this MTP, large-scale MD simulations at elevated temperatures reveal atomic-scale interfacial reactions, including the breakdown of S_8 rings into smaller S_x clusters and Li- S_x species, and quantify the resulting Li^+ diffusion barriers. The ML-driven simulations identify that while interfacial reactions tend to increase activation energies, the dominant factor controlling Li^+ transport is the surface-specific diffusion topology of $\beta\text{-Li}_3\text{PS}_4$, with the (100) surface offering favorable 2D diffusion channels. This approach demonstrates how ML potentials can bridge the gap between DFT accuracy and large-scale MD, enabling predictive insights into interfacial stability and ion kinetics that are otherwise inaccessible with conventional ab initio methods.

Extending beyond specific material systems, Devereaux and colleagues developed a low-cost, data-driven computational model that predicted the reactivity at interfaces between candidate SSEs and lithium metal anodes using atomistic structural information [85]. The model was trained on AIMD simulations data of the time evolution of the solid electrolyte-Li metal interfaces for 67 different materials. Structural features (e.g., average atomic volume, packing fraction, sublattice bond ionicity) along with thermodynamic properties (DFT formation energy E_{form} and ΔE_{rxn} with Li metal) were used as descriptors. Using this training set, two logistic regression models were built to classify the reactivity of SSE candidates with Li metal: a “reactive model” to differentiate between stable and reactive/passivating materials, and an “unreactive model” to differentiate between stable/passivating and reactive materials. The framework identified 533 stable and 788 passivating material interfaces, offering an expanded set of candidates for subsequent discovery and screening efforts in SSBs (Fig. 9). This study represents a pioneering attempt to systematically classify interfacial reactivity across diverse chemical spaces for SSBs, and its methodology can be readily extended to evaluate reactivity in other solid-solid systems.

In addition to sulfide-based electrolytes, garnet-type SSEs also face critical interfacial challenges. Wan et al. leverages a ML interatomic potential to perform large-scale MD simulations, enabling atomic-scale investigation into the initial degradation mechanisms at the interface between garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ SSEs and LiCoO_2 (LCO) cathode [86]. The MLIP was trained on extensive ab initio data covering crystalline and amorphous phases of LLZO and LCO, as well as mixed and Co-doped structures. The developed ML model can perform MD simulations with low energy and force errors and predict structural, vibrational, and dynamical properties of LLZO/LCO mixture models as well as bulk LLZO and LCO with similar accuracy to DFT. This work elaborated the fundamental relationship between atomic structure, chemistry, and transport behavior at complex LLZO interfaces with the aid of a flexible and accurate ML interatomic potential.

Beyond structural and chemical stability, the kinetics of interphase formation are equally critical. Yang et al. employed thermodynamic phase equilibrium analysis and ML-potential-assisted MD simulations to elucidate the thermodynamics and kinetics of SEI formation at the Li/ $\beta\text{-Li}_3\text{PS}_4$ interface at the atomic scale [87]. An accurate moment tensor potential using the ML method was developed for a reactive model of Li/ $\beta\text{-Li}_3\text{PS}_4$. This potential enabled large-scale MD simulations with the model size expanded to the experimental dimensions (~ 40 nm) while maintaining accuracy comparable to density functional theory calculations. The ML-assisted simulations revealed the SEI formation process in four stages: fast ion diffusion, nucleation, Li_2S growth, and stabilization (Fig. 10).

Considering the promising role of SSBs in high-temperature environments, mechanical instability within the cell during operation is

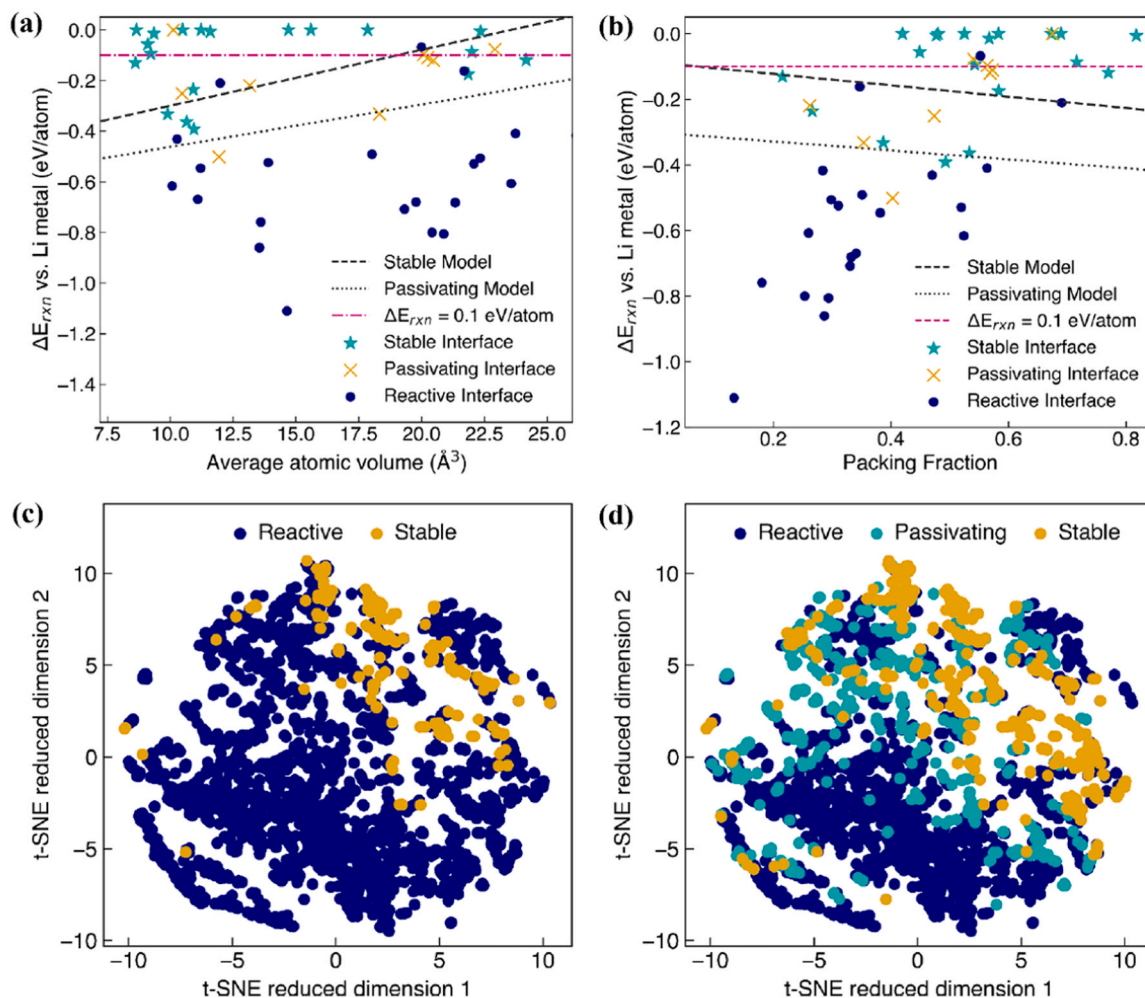


Fig. 9. Decision boundary for the 2-feature reactivity predictors and resulting expansion of the material search space. Top two panels show the spread of the training data in 2-feature space. In purple we show the decision boundary of the current thermodynamic model -100 meV/atom. The decision boundaries of the 2-feature SM and RM type binary predictors, based on different splits of the training data, which represent the kinetic stabilization barrier dependence on the average atomic volume (a) and packing fraction (b). The lower two panels depict 3535 SSE candidates in the t-SNE representation of our full feature space. Panel (c) shows the labeling using the current method of interfacial reactivity prediction, where all interfaces with $|\Delta E_{rxn}| < 100$ meV/atom are considered stable, totaling 229 candidates. In panel (d) we present the predictions for the combined SM and RM framework, with 533 predicted stable interfaces and 788 predicted passivating interfaces [85].

anticipated due to inadequate thermomechanical compatibility among the solid battery components, including electrodes and electrolytes. This compatibility is governed by multiple intrinsic thermal characteristics of the materials, such as thermal conductivity, thermal stability, and thermal expansion. Such mismatches induce thermal stresses, which can ultimately result in system failure under prolonged operational conditions. Notably, mismatched thermal expansion across battery materials resulting from poor thermomechanical compatibility may lead to mechanical degradation, reduce charge storage capacity, and create safety risks. To tackle this, Yoon et al. compiled an open-source dataset of coefficients of thermal expansion (CTE) to train ML models [88]. A total of 132 compositional attributes were generated for each compound by processing 22 electronic, physical, and chemical properties with six statistical operations: average, maximum, standard deviation, minimum, range, and mode. These attributes were further enhanced by integrating 139 crystal structure-based features, culminating in a comprehensive set of 271 composition and crystal features for every material. The validated model was then deployed for high-throughput screening, predicting CTE values for 25,819 compounds from the ICSD database. From these predictions, the researchers systematically evaluated 756 cathode-electrolyte combinations based on two key

compatibility metrics: the difference in their CTE values (to minimize thermal mismatch) and their average CTE (to reduce overall thermal expansion). This dual-criterion approach identified several promising material pairs, such as LiMnO_4P paired with $\text{Li}_2\text{S}_2\text{V}$, which exhibited both minimal CTE difference and low average expansion.

These studies collectively illustrate that ML, especially through the creation of precise interatomic potentials and data-driven classification models, offers a transformative set of tools for understanding and designing stable interfaces in SSBs. By facilitating large-scale, high-fidelity simulations and enabling systematic predictions of interfacial reactivity, ML methods tackle essential challenges such as chemical reactions, space-charge effects, degradation kinetics, solid electrolyte interphase formation, and thermomechanical compatibility—all critical to the performance and durability of SSBs. This integrated computational approach not only unravels complex interfacial phenomena but also accelerates the discovery and optimization of materials with compatible interfaces, thereby paving the way for more reliable and high energy density SSBs.

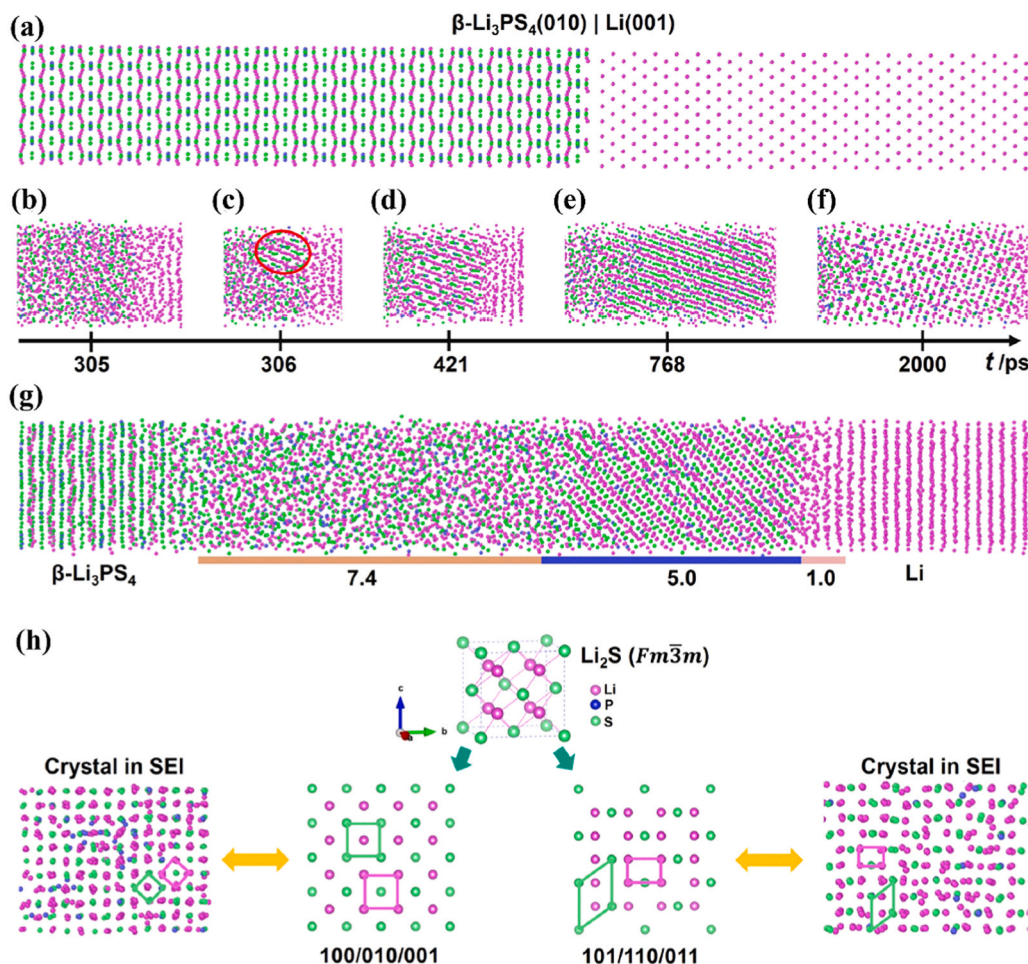


Fig. 10. SEI layer formation process between Li (001) and $\beta\text{-Li}_3\text{PS}_4$ (010) at 300 K and 100 bar. (a) Initial interface model, (b) ion diffusion, (c) and (d) nucleation and (e) crystal growth, and (f) stabilization, and (g) final structure of the reactive interface model. (h) Structure of the crystalline SEI region along different directions and surface structure of Li_2S with space group $Fm\bar{3}m$ along 101/110/011 and 100/010/001 [87].

6. Challenges, prospects, and conclusion

Despite the remarkable progress in applying ML to SSEs research, several critical challenges persist. Addressing these challenges will be essential for fully realizing the transformative potential of data-driven approaches in accelerating the development of high-performance SSBs.

- (1) Data quality and standardization:** The efficacy of ML models fundamentally depends on the quality, quantity, and consistency of training data. Based on the comprehensive framework for governing machine learning-oriented materials data quality [89], "AI-Ready" material data in the context of SSEs must satisfy a multi-dimensional quality standard that integrates both structured property values and rich, domain-specific metadata. Specifically, such data should include not only numerical descriptors (e.g., ionic conductivity, migration barriers, formation energies, and electrochemical stability windows) but also systematically documented metadata covering computational parameters (e.g., DFT exchange-correlation functionals, k-point sampling, convergence thresholds, pseudopotential types, and thermodynamic ensembles used in molecular dynamics simulations) and experimental conditions (e.g., synthesis routes, sintering temperatures, atmospheric controls, purity levels of precursors, annealing protocols, and characterization techniques). Moreover, the data should adhere to standardized formats (e.g., CIF for crystal structures) and be governed through a lifecycle-aware

quality model that ensures traceability (provenance of data sources and processing steps), consistency (uniform representation across multi-source datasets), completeness (no missing values or critical information), and redundancy control (elimination of duplicated or non-informative features). The framework emphasizes the integration of materials domain knowledge to validate data plausibility, identify outliers based on physico-chemical constraints, and guide feature selection. For instance, in SSE datasets, metadata on phase stability, impurity phases, and electrode compatibility should be recorded to contextualize property predictions. Only through such rigorous, metadata-enriched, and domain-informed data governance can machine learning models achieve reliable, interpretable, and transferable performance. Efforts toward standardized and open data reporting for SSEs are being advanced by several key platforms. These include dedicated resources such as the EESMDP [60], DDSE [90], and the Inorganic Solid Electrolyte Materials Database [91], as well as comprehensive materials databases like the Materials Project.

Currently, datasets are often fragmented, derived from diverse sources (experimental measurements, DFT calculations, literature mining) with inconsistent formats, varying measurement conditions, and missing uncertainty quantification. This heterogeneity limits model generalizability and reproducibility. Furthermore, a prevalent publication bias towards reporting successful materials leads to imbalanced datasets lacking

information on failed syntheses or suboptimal properties, which are crucial for learning complex property trade-offs. To mitigate this data imbalance, several algorithmic approaches can be implemented. **Active Learning:** Iterative querying of high-uncertainty regions in the material space, followed by targeted experiments or simulations, to efficiently expand the dataset with informative "negative" examples. **Synthetic Data Generation:** Techniques such as Synthetic Minority Over-sampling Technique (SMOTE) and Generative Adversarial Networks (GANs) can artificially augment underrepresented classes (e.g., low-conductivity or unstable materials), though careful validation is required to ensure physical plausibility. **Transfer Learning:** Leveraging pre-trained models on large, general materials databases (e.g., Materials Project) and fine-tuning them on smaller, domain-specific SSE datasets, thereby reducing the dependency on extensive labeled data. **Failure-Inclusive Databases:** Encouraging community initiatives to systematically archive and share "unsuccessful" experimental outcomes, along with contextual metadata, to build more balanced and realistic training sets. These approaches not only improve model robustness but also accelerate the discovery of materials with optimized trade-offs between multiple properties. Future efforts must prioritize the establishment of open, standardized, and multi-modal databases for SSEs. These databases should encompass not only structural and property data but also detailed metadata on synthesis conditions, characterization methods, and error estimates. Community-wide initiatives and standardized data reporting protocols are needed to foster data sharing and build comprehensive, high-fidelity datasets that can support robust and generalizable ML models.

Furthermore, a substantial gap still exists between ML predictions and experimental synthesis. Several common bottlenecks hinder the translation of computationally predicted materials into experimentally realizable SSEs. First, many ML models are trained on idealized structural data and equilibrium properties, while kinetic limitations, impurity formation, and interfacial reactions that inevitably occur during real synthesis are often neglected. Second, the diversity and uncertainty of synthesis pathways, including precursor selection, reaction routes, and processing conditions, make it challenging to experimentally access ML predicted materials using conventional synthesis methods. Third, inconsistencies between the parameters assumed in computational models, such as temperature, pressure, or defect concentrations, and actual experimental conditions frequently lead to discrepancies between predicted properties and measured performance. To improve the practical applicability of ML in SSEs design, future research should emphasize the development of an integrated computation, synthesis, and characterization closed loop framework. In particular, active learning strategies that iteratively incorporate experimental feedback to optimize synthesis parameters are highly desirable. In parallel, establishing standardized and open databases that systematically record synthesis conditions, processing protocols, and post treatment information will be essential for accelerating the transition from virtual materials design to experimental realization.

- (2) **Multi-Properties prediction:** In the development of SSEs, multi-property collaborative optimization presents a core challenge, requiring a delicate balance among multiple key properties to meet the specific requirements of different application scenarios. Taking the relationship between elastic modulus and ionic conductivity as an example: an increase in the former often leads to a certain degree of reduction in the latter, which stems from the inherent physical constraints between material mechanical stiffness and ion migration capability within the lattice. Therefore, to maintain efficient ion transport while ensuring structural mechanical integrity, an optimal balance must be struck between the

two, thereby achieving the overall performance optimization of the battery. To address this challenge, constructing a high-quality training dataset is crucial. The data should not only encompass "ideal materials" with balanced properties but also systematically include "non-optimal materials" that excel in certain aspects while exhibiting shortcomings in others. Such data can comprehensively depict the constraints between properties, providing a foundation for ML models to learn the complex trade-offs among performance characteristics. In terms of model construction, it is advisable to develop a unified model capable of simultaneously predicting multiple key property parameters (such as elastic modulus, ionic conductivity, migration barriers, etc.). Compared to training multiple independent models, this multi-task learning framework can more effectively capture the inherent coupling and mutual constraints among properties. This enables a more accurate quantification of the trade-off intensity between different properties and facilitates the precise identification and design of optimal material combinations tailored to specific application scenarios.

In fact, the effective implementation of the aforementioned multi-task learning framework, especially when dealing with kinetic properties such as interfacial ion migration barriers, is often constrained by the scarcity of relevant high-quality data. To address this, a viable solution is to adopt transfer learning strategies [92]. Transfer learning offers an effective route to address the severe data scarcity associated with kinetic properties in battery materials. In this framework, ML models are first pre-trained on large-scale general-purpose materials databases (e.g., the Materials Project) containing abundant structural, energetic, and stability data, enabling the learning of transferable representations of local atomic environments, bonding characteristics, and geometric motifs. These pretrained models are then adapted to target tasks involving limited data, such as interfacial ion migration barriers and reaction kinetics, via hierarchical fine-tuning, in which lower-level structural encoders are frozen or weakly updated, while higher-level interaction layers and task-specific prediction heads are selectively optimized. The fine-tuning process relies on a small set of high-fidelity first-principles data, including nudged elastic band migration barriers, adsorption energies, and initial-final state energy differences, and can be further enhanced through layer-wise unfreezing or multi-task learning strategies. This transfer-learning paradigm substantially reduces the dependence on large kinetic datasets while maintaining high predictive accuracy, thereby enabling scalable and generalizable modeling of complex interfacial kinetics. More importantly, when combined with multi-task learning, it provides a practical framework for quantitatively resolving property trade-offs and guiding data-driven screening of SSEs under realistic data limitations.

- (3) **Interpretability of ML models:** The application of ML in SSEs research is evolving from traditional data-driven prediction toward an interpretable intelligent design paradigm that integrates physical mechanisms and domain knowledge [93]. Current challenges include: the scarcity of high-quality data coupled with publication bias, which hinders models from learning a comprehensive map of material behavior; the disconnection between the predictions of black-box models and underlying physical mechanisms, preventing the extraction of deep principles to guide design; and the difficulty in transforming experts' qualitative knowledge into computable constraints. Future efforts will focus on the following key directions: developing physics-informed ML architectures that encode fundamental physical laws and symmetry constraints into model design to enhance reliability and interpretability; constructing human-machine collaborative active learning closed-loop platforms that combine experts' qualitative judgment with machines' massive search capabilities

to achieve efficient and rational material exploration; and leveraging explainable AI tools to inversely extract key physical descriptors, reducing reliance on manually crafted features and promoting the discovery of new scientific laws.

- (4) **Generative design:** While ML has demonstrated remarkable prowess in predicting the properties of SSEs, a transformative next frontier lies in transitioning from a descriptive, screening-based paradigm to a generative and design-centric one. The ultimate goal is not merely to identify promising candidates from a pre-defined set, but to invert the design process: to encode a set of target performance metrics and directly generate novel, chemically viable material compositions and structures that satisfy these multi-objective constraints. Realizing this vision necessitates overcoming three major barriers: the high-dimensional and complex design space requiring multi-objective trade-offs, the disconnect between data-driven models and underlying physicochemical mechanisms, and the feasibility gap bridging virtual design with experimental synthesis. Here, beyond traditional high-throughput screening, recent studies have begun to apply generative machine-learning models directly to the discovery of SSEs candidates. Notably, Nguyen and co-workers introduced a physics-informed hierarchical generative framework that integrates symmetry-aware crystallographic principles with empirical constraints to systematically explore lattice parameters, bonding environments, and packing densities for SSEs [94]. This approach generated chemically valid and structurally stable candidate phases with high ionic conductivity, demonstrating the potential of generative design in SSEs discovery. In addition, general deep generative models have been shown to generate novel crystalline materials beyond databases, suggesting broader opportunities for inverse design of functional materials [95–99]. Future research must focus on developing generative models that integrate physical rules, constructing active learning closed loops that connect computational design with automated synthesis, and establishing standardized databases to support inverse design. Driving the paradigm shift of ML from a "predictive tool" to a "design engine" will be the key to unlocking the on-demand customization of high-performance SSEs and ultimately enabling the rational design of SSBs.

- (5) **Holistic battery optimization:** Our ultimate goal is not merely a high-performance SSE, but rather an exceptional full battery system. This shifts the optimization target from the material itself to a complex system encompassing multi-scale, multi-physics coupling. Full battery performance spans physical processes across atomic scales (interfacial reaction energy barriers), mesoscopic scales (ion/electron transport networks), and macroscopic scales (thermo-mechanical stress distributions). However, computational simulations and experimental characterization data from different sources vary significantly in format, precision, and spatiotemporal dimensions, creating fundamental difficulties in constructing a unified, coherent multi-scale dataset. The system design objectives represent a multidimensional, dynamically evolving vector of often mutually constraining metrics, including energy density, power density, cycle life, safety, and cost. For instance, pursuing higher energy density may compromise safety margins, and such trade-off relationships further evolve in real-time with operating conditions. Traditional single-objective regression or classification models are inadequate for such problems, urgently necessitating the development of intelligent optimization and decision frameworks capable of handling conflicting objectives and adapting to dynamic scenarios. Future ML models must transcend single-property prediction to achieve collaborative modeling and optimization across electrochemistry, thermal, and mechanical physics. Only by doing so can they accurately reflect the complex behavior of batteries under real-world operating conditions, thereby truly

advancing full battery systems from being merely "functional" to "exceptional."

In conclusion, while ML has already begun to reshape the paradigm of SSEs research, overcoming the aforementioned challenges will require concerted efforts from the communities of materials science, electrochemistry and computer science. By fostering interdisciplinary collaboration, investing in infrastructure for data and automation, and developing more intelligent, interpretable, and integrated ML frameworks, we can accelerate the journey toward high-performance, safe, and commercially viable SSBs.

CRediT authorship contribution statement

Jun Ma: Writing – original draft. **Xiang Yin:** Writing – original draft. **Guanglei Cui:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Shu Zhang:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was financially supported by the National Key R&D Program of China (2023YFC2812700), Natural Science Foundation of Shandong Province (ZR2025MS177), and the National Natural Science Foundation of China (22139001).

Data availability

No data was used for the research described in the article.

References

- [1] B. Scrosati, J. Hassoun, Y.-K. Sun, Lithium-ion batteries. A look into the future, *Energy Environ. Sci.* 4 (9) (2011) 3287–3295, <https://doi.org/10.1039/c1ee01388b>.
- [2] M.M. Thackeray, C. Wolverton, E.D. Isaacs, Electrical energy storage for transportation: approaching the limits of, and going beyond, lithium-ion batteries, *Energy Environ. Sci.* 5 (7) (2012) 7854–7863, <https://doi.org/10.1039/C2EE21892E>.
- [3] D. Lin, Y. Liu, Y. Cui, Reviving the lithium metal anode for high-energy batteries, *Nat. Nanotechnol.* 12 (3) (2017) 194–206, <https://doi.org/10.1038/nnano.2017.16>.
- [4] X.B. Cheng, R. Zhang, C.Z. Zhao, Q. Zhang, Toward safe lithium metal anode in rechargeable batteries: a review, *Chem. Rev.* 117 (15) (2017) 10403–10473, <https://doi.org/10.1021/acs.chemrev.7b00115>.
- [5] K.B. Hatzell, X.C. Chen, C.L. Cobb, N.P. Dasgupta, M.B. Dixit, L.E. Marbella, M. T. McDowell, P.P. Mukherjee, A. Verma, V. Viswanathan, et al., Challenges in lithium metal anodes for solid-state batteries, *ACS Energy Lett.* 5 (3) (2020) 922–934, <https://doi.org/10.1021/acsenergylett.9b02668>.
- [6] Z. Zhang, W.Q. Han, From liquid to solid-state lithium metal batteries: fundamental issues and recent developments, *Nanomicro Lett.* 16 (1) (2023) 24, <https://doi.org/10.1007/s40820-023-01234-y>.
- [7] S. Raza, T. Bashir, A. Hayat, H.S.M. Abd-Rabboh, L. Shen, Y. Orooji, H. Lin, Recent progress and fundamentals of solid-state electrolytes for all solid-state rechargeable batteries: mechanisms, challenges, and applications, *J. Energy Storage* 92 (2024), <https://doi.org/10.1016/j.est.2024.112110>.
- [8] J. Janek, W.G. Zeier, Challenges in speeding up solid-state battery development, *Nat. Energy* 8 (3) (2023) 230–240, <https://doi.org/10.1038/s41560-023-01208-9>.
- [9] C. Wang, J.T. Kim, C. Wang, X. Sun, Progress and prospects of inorganic solid-state electrolyte-based all-solid-state pouch cells, *Adv. Mater.* 35 (19) (2023) e2209074, <https://doi.org/10.1002/adma.202209074>.
- [10] Z. Zheng, J. Zhou, Y. Zhu, Computational approach inspired advancements of solid-state electrolytes for lithium secondary batteries: from first-principles to machine learning, *Chem. Soc. Rev.* 53 (6) (2024) 3134–3166, <https://doi.org/10.1039/d3cs00572k>.

- [11] Y. Zhu, C. Wang, D. Guo, X. Chen, S. Wang, Solid-state electrolytes: probing interface regulation from multiple perspectives, *ACS Appl. Mater. Interfaces* 16 (33) (2024) 43114–43133, <https://doi.org/10.1021/acsami.4c07428>.
- [12] C. Xu, L. Chen, F. Wu, Unveiling the power of sulfide solid electrolytes for next-generation all-solid-state lithium batteries, *Materials* 6 (2025) 100428, <https://doi.org/10.1016/j.nxmate.2024.100428>.
- [13] Y. Pilyugina, E.V. Kuzmina, V.S. Kolosnitsyn, Review—recent advancements in sulfide solid electrolytes for all-solid-state lithium-sulfur batteries, *ECS J. Solid State Sci. Technol.* 13 (6) (2024), <https://doi.org/10.1149/2162-8777/ad575f>.
- [14] S. Yang, Y. Tang, Y. Yao, S. He, Z. Wu, Y. Yang, H. Pan, X. Rui, Y. Yu, Sulfide electrolytes for all-solid-state sodium batteries: fundamentals and modification strategies, *Mater. Horiz.* 12 (4) (2025) 1058–1083, <https://doi.org/10.1039/d4mh01218f>.
- [15] S. Ai, X. Wu, J. Wang, X. Li, X. Hao, Y. Meng, Research progress on solid-state electrolytes in solid-state lithium batteries: classification, ionic conductive mechanism, interfacial challenges, *Nanomater. (Basel)* 14 (22) (2024), <https://doi.org/10.3390/nano14221773>.
- [16] Z. Karkar, M.S.E. Houache, C.-H. Yim, Y. Abu-Lebdeh, An industrial perspective and intellectual property landscape on solid-state battery technology with a focus on solid-state electrolyte chemistries, *Batteries* 10 (1) (2024), <https://doi.org/10.3390/batteries10010024>.
- [17] J.-H. Yin, H. Zhu, S.-J. Yu, Y.-B. Dong, Q.-Y. Wei, G.-Q. Xu, Y. Xiong, Y. Qian, Recent advances of LATP and their NASICON structure as a solid-state electrolyte for lithium-ion batteries, *Adv. Eng. Mater.* 25 (20) (2023), <https://doi.org/10.1002/adem.202300566>.
- [18] S. Zhang, J. Ma, S. Dong, G. Cui, Designing all-solid-state batteries by theoretical computation: a review, *Electrochem. Energy Rev.* 6 (1) (2023), <https://doi.org/10.1007/s41918-022-00143-9>.
- [19] L. Xia, H. Liu, Y. Pei, Theoretical calculations and simulations power the design of inorganic solid-state electrolytes, *Nanoscale* 16 (33) (2024) 15481–15501, <https://doi.org/10.1039/d4nr02114b>.
- [20] M. Wu, Z. Wei, Y. Zhao, Q. He, Recent applications of theoretical calculations and artificial intelligence in solid-state electrolyte research: a review, *Nanomaterials* 15 (3) (2025), <https://doi.org/10.3390/nano15030225>.
- [21] A. Agrawal, A. Choudhary, Perspective: Materials informatics and big data: realization of the “fourth paradigm” of science in materials science, *APL Mater.* 4 (5) (2016), <https://doi.org/10.1063/1.4946894>.
- [22] R. Jose, S. Ramakrishna, Materials 4.0: Materials big data enabled materials discovery, *Appl. Mater. Today* 10 (2018) 127–132, <https://doi.org/10.1016/j.apmt.2017.12.015>.
- [23] T. Lombardo, M. Duquesnoy, H. El-Bouysidy, F. Aren, A. Gallo-Bueno, P. B. Jorgensen, A. Bhowmik, A. Demortiere, E. Ayerbe, F. Alcaide, et al., Artificial intelligence applied to battery research: hype or reality? *Chem. Rev.* (2021) <https://doi.org/10.1021/acs.chemrev.1c00108>.
- [24] B. Mortazavi, Recent advances in machine learning-assisted multiscale design of energy materials, *Adv. Energy Mater.* 15 (9) (2024), <https://doi.org/10.1002/aenm.202403876>.
- [25] G. Xu, M. Jiang, J. Li, X. Xuan, J. Li, T. Lu, L. Pan, Machine learning-accelerated discovery and design of electrode materials and electrolytes for lithium ion batteries, *Energy Stor. Mater.* 72 (2024), <https://doi.org/10.1016/j.ensm.2024.103710>.
- [26] R. Li, W. Zhao, R. Li, C. Gan, L. Chen, Z. Wang, X. Yang, Leveraging machine learning for accelerated materials innovation in lithium-ion battery: a review, *J. Energy Chem.* 106 (2025) 44–62, <https://doi.org/10.1016/j.jechem.2025.02.038>.
- [27] Y. Wang, Application-oriented design of machine learning paradigms for battery science, *NPJ Comput. Mater.* 11 (1) (2025), <https://doi.org/10.1038/s41524-025-01575-9>.
- [28] Y. Liu, Q. Zhou, G. Cui, Machine learning boosting the development of advanced lithium batteries, *Small Methods* 5 (8) (2021) e2100442, <https://doi.org/10.1002/smt.202100442>.
- [29] R. Xiao, H. Li, L. Chen, High-throughput design and optimization of fast lithium ion conductors by the combination of bond-valence method and density functional theory (14221–14211), *Sci. Rep.* 5 (2015) 14227, <https://doi.org/10.1038/srep14227>.
- [30] B. Liu, D. Wang, M. Avdeev, S. Shi, J. Yang, W. Zhang, High-throughput computational screening of Li-containing fluorides for battery cathode coatings, *ACS Sustain. Chem. Eng.* 8 (2) (2019) 948–957, <https://doi.org/10.1021/acssuschemeng.9b05557>.
- [31] S. Mui, J. Voss, R. Schlem, R. Koerver, S.J. Sedlmaier, F. Maglia, P. Lamp, W. G. Zeier, Y. Shao-Horn, High-throughput screening of solid-state Li-ion conductors using lattice-dynamics descriptors, *iScience* 16 (2019) 270–282, <https://doi.org/10.1016/j.isci.2019.05.036>.
- [32] L. Kahle, A. Marcolongo, N. Marzari, High-throughput computational screening for solid-state Li-ion conductors, *Energy Environ. Sci.* 13 (3) (2020) 928–948, <https://doi.org/10.1039/c9ee02457c>.
- [33] M. Liu, A. Gopakumar, V. Hegde, J. He, C. Wolverton, High-throughput hybrid-functional DFT calculations of bandgaps and formation energies and multifidelity learning with uncertainty quantification, *Phys. Rev. Mater.* 8 (4) (2024), <https://doi.org/10.1103/PhysRevMaterials.8.043803>.
- [34] W. Chen, Y. Zhao, J. Zhou, S. Li, C. Lu, S. Zhou, H. Li, Y. Li, Y. Cheng, J. Yang, et al., High-throughput screening of halide solid-state electrolytes for all-solid-state Li-ion batteries through structural descriptor, *J. Alloy. Compd.* (2025) 1010, <https://doi.org/10.1016/j.jallcom.2024.177167>.
- [35] C. Lin, L. Zhang, Y. Dong, Compositional machine learning and high-throughput screening aided discovery of novel anti-perovskite solid-state electrolytes, *J. Energy Storage* 125 (2025), <https://doi.org/10.1016/j.jest.2025.116990>.
- [36] Z. Xu, Y. Xia, H. Duan, J. Li, Y. Lin, Y. Jiang, X. Feng, Z. Liu, M. Zheng, Y. Xia, High-throughput calculations and machine learning for discovering halide-type solid-state electrolyte materials, *Chem. Mater.* 37 (17) (2025) 6642–6654, <https://doi.org/10.1021/acs.chemmater.5c01169>.
- [37] J. Kim, D. Lee, D. Lee, X. Li, Y.-L. Lee, S. Kim, Machine learning prediction models for solid electrolytes based on lattice dynamics properties, *J. Phys. Chem. Lett.* 15 (22) (2024) 5914–5922, <https://doi.org/10.1021/acs.jpclett.4c00995>.
- [38] L. Tang, G. Zhang, J. Jiang, Machine learning approach accelerates search for solid state electrolytes, *Chin. J. Chem. Phys.* 37 (4) (2024) 505–512, <https://doi.org/10.1063/1674-0068/cjcp2402020>.
- [39] A.C.C. Dutra, B.A. Goldmann, M.S. Islam, J.A. Dawson, Understanding solid-state battery electrolytes using atomistic modelling and machine learning, *Nat. Rev. Mater.* 10 (8) (2025) 566–583, <https://doi.org/10.1038/s41578-025-00817-y>.
- [40] T. Gao, Y. Wu, Applications and advances of machine learning in the development of solid-state electrolytes for lithium-ion batteries, *ACS Omega* (2025), <https://doi.org/10.1021/acsomega.5c08467>.
- [41] V. Jain, Z. Wang, F. You, Machine learning pipelines for the design of solid-state electrolytes, *Mater. Horiz.* (2026), <https://doi.org/10.1039/d5mh01525a>.
- [42] J. Li, M. Zhou, H.H. Wu, L. Wang, J. Zhang, N. Wu, K. Pan, G. Liu, Y. Zhang, J. Han, et al., Machine learning-assisted property prediction of solid-state electrolyte, *Adv. Energy Mater.* 14 (20) (2024), <https://doi.org/10.1002/aenm.202304480>.
- [43] K. Kim, D.J. Siegel, Machine learning reveals factors that control ion mobility in anti-perovskite solid electrolytes, *J. Mater. Chem. A* 10 (28) (2022) 15169–15182, <https://doi.org/10.1039/d2ta03613d>.
- [44] D. Rajagopal, A. Cierpka, B. Nestler, A. Koeppel, Generative deep learning for advanced battery materials, *Batter. & Supercaps* (2025), <https://doi.org/10.1002/batt.202500494>.
- [45] M.A. Tamer, X. Lin, J. a Wang, L. Cai, A. Lahmar, J. Ma, M. Yang, Accelerated discovery of solid-state battery properties enabled by active learning approaches, *Sci. China Mater.* 68 (6) (2025) 1974–1985, <https://doi.org/10.1007/s40843-025-3346-4>.
- [46] M.H. Müser, S.V. Sukhomlinov, L. Pastewka, Interatomic potentials: achievements and challenges, *Adv. Phys. X* 8 (1) (2022), <https://doi.org/10.1080/23746149.2022.2093129>.
- [47] X. He, Y. Zhu, A. Epstein, Y. Mo, Statistical variances of diffusional properties from ab initio molecular dynamics simulations, *NPJ Comput. Mater.* 4 (1) (2018) 65–73, <https://doi.org/10.1038/s41524-018-0074-y>.
- [48] F. Mouvet, J. Villard, V. Bolnykh, U. Rothlisberger, Recent advances in first-principles based molecular dynamics, *Acc. Chem. Res.* 55 (3) (2022) 221–230, <https://doi.org/10.1021/acs.accounts.1c00503>.
- [49] V.L. Deringer, M.A. Caro, G. Csanyi, Machine learning interatomic potentials as emerging tools for materials science, *Adv. Mater.* 31 (46) (2019) e1902765, <https://doi.org/10.1002/adma.201902765>.
- [50] J. Chen, M. Fang, Q. Wu, S. Tang, J. Zheng, C. Wei, X. Cao, Y. Shi, N. Xu, Y. He, Insights into the Atomic Mechanism of Lithium-Ion Diffusion in Li6PS5Cl via a Machine Learning Potential, *Chem. Mater.* 37 (2) (2025) 591–599, <https://doi.org/10.1021/acs.chemmater.4c01152>.
- [51] S. Park, J. Kang, Atomistic insights into room-temperature ion conduction mechanisms in Li10GeP2S12 via machine learning interatomic potentials, *J. Appl. Phys.* 138 (13) (2025), <https://doi.org/10.1063/5.0293554>.
- [52] R. Wang, M. Guo, Y. Gao, X. Wang, Y. Zhang, B. Deng, M. Shi, L. Zhang, Z. Zhong, A pre-trained deep potential model for sulfide solid electrolytes with broad coverage and high accuracy, *NPJ Comput. Mater.* 11 (1) (2025), <https://doi.org/10.1038/s41524-025-01764-6>.
- [53] Q. Chen, S. Wang, C. Ling, Redefining atomistic simulations of all-solid-state batteries through machine learning interatomic potentials, *J. Energy Chem.* 112 (2026) 666–687, <https://doi.org/10.1016/j.jechem.2025.08.058>.
- [54] P. Xue, R. Qiu, C. Peng, Z. Peng, K. Ding, R. Long, L. Ma, Q. Zheng, Solutions for lithium battery materials data issues in machine learning: overview and future outlook, *Adv. Sci. (Weinh.)* 11 (48) (2024) e2410065, <https://doi.org/10.1002/advs.202410065>.
- [55] H. Liu, S. Li, S. Zhu, Y. Hu, X. Han, C. Shi, F. He, C. He, B. Chen, N. Zhao, How machine learning has driven the development of rechargeable ion batteries, *Adv. Energy Mater.* 15 (47) (2025), <https://doi.org/10.1002/aenm.202504095>.
- [56] B. Ouyang, J. Wang, T. He, C.J. Bartel, H. Huo, Y. Wang, V. Laciuta, H. Kim, G. Ceder, Synthetic accessibility and stability rules of NASICONs, *Nat. Commun.* 12 (1) (2021), <https://doi.org/10.1038/s41467-021-26006-3>.
- [57] R. Arróyave, Phase stability through machine learning, *J. Phase Equilibria Diffus.* 43 (6) (2022) 606–628, <https://doi.org/10.1007/s11669-022-01009-9>.
- [58] C.J. Bartel, A. Trewartha, Q. Wang, A. Dunn, A. Jain, G. Ceder, A critical examination of compound stability predictions from machine-learned formation energies, *NPJ Comput. Mater.* 6 (1) (2020), <https://doi.org/10.1038/s41524-020-00362-y>.
- [59] C. Bai, Y. Li, G. Xiao, J. Chen, S. Tan, P. Shi, T. Hou, M. Liu, Y.B. He, F. Kang, Understanding the electrochemical window of solid-state electrolyte in full battery application, *Chem. Rev.* 125 (14) (2025) 6541–6608, <https://doi.org/10.1021/acs.chemrev.4c01012>.
- [60] X. Wang, B. He, B. Liu, M. Avdeev, S. Shi, A database of electrochemical stability windows containing over 1500 solid-state inorganic compounds, *Adv. Funct. Mater.* 34 (44) (2024), <https://doi.org/10.1002/adfm.202406146>.
- [61] J. Chen, L. Jiang, S. Tan, J. Yang, Z. Li, C. Bai, X. Zhang, R. Li, Y. Xie, M. Liu, et al., Machine-learning-aided screening of inorganic lithium solid-state electrolytes with

- a wide electrochemical window, *J. Mater. Chem. A* 13 (29) (2025) 23445–23453, <https://doi.org/10.1039/d5ta04197j>.
- [62] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized Gradient Approximation Made Simple, *Phys. Rev. Lett.* 77 (18) (1996) 3865.
- [63] Z. Wang, X. Lin, Y. Han, J. Cai, S. Wu, X. Yu, J. Li, Harnessing artificial intelligence to holistic design and identification for solid electrolytes, *Nano Energy* 89 (2021), <https://doi.org/10.1016/j.nanoen.2021.106337>.
- [64] A.D. Sendek, Q. Yang, E.D. Cubuk, K.-A.N. Duerloo, Y. Cui, E.J. Reed, Holistic computational structure screening of more than 12 000 candidates for solid lithium-ion conductor materials, *Energy Environ. Sci.* 10 (1) (2017) 306–320, <https://doi.org/10.1039/c6ee02697d>.
- [65] Y. Zhang, X. He, Z. Chen, Q. Bai, A.M. Nolan, C.A. Roberts, D. Banerjee, T. Matsunaga, Y. Mo, C. Ling, Unsupervised discovery of solid-state lithium ion conductors, *Nat. Commun.* 10 (1) (2019) 5260, <https://doi.org/10.1038/s41467-019-13214-1>.
- [66] C.J. Hargreaves, M.W. Gaultois, L.M. Daniels, E.J. Watts, V.A. Kurlin, M. Moran, Y. Dang, R. Morris, A. Morscher, K. Thompson, et al., A database of experimentally measured lithium solid electrolyte conductivities evaluated with machine learning, *NPJ Comput. Mater.* 9 (1) (2023), <https://doi.org/10.1038/s41524-022-00951-z>.
- [67] B. He, A. Ye, S. Chi, P. Mi, Y. Ran, L. Zhang, X. Zou, B. Pu, Q. Zhao, Z. Zou, et al., CAVD, towards better characterization of void space for ionic transport analysis, *Sci. Data* 7 (1) (2020), <https://doi.org/10.1038/s41597-020-0491-x>.
- [68] Bing He, A. Y. Chi Shuting, Penghui Mi, Yunbing Ran, Liwen Zhang, Xinxin Zou, Bowei Pu, Qian Zhao, Zheyi Zou, Da Wang, Wenqing Zhang, Jingtai Zhao, Maxim Avdeev, Siqu Shi, Ion-transport descriptor data for 6,995 Li-, Na-, Mg-, and Al-containing compounds, *Sci. Data* (2019), <https://doi.org/10.6084/m9.figshare.11448159.v4>.
- [69] B. He, S. Chi, A. Ye, P. Mi, L. Zhang, B. Pu, Z. Zou, Y. Ran, Q. Zhao, D. Wang, et al., High-throughput screening platform for solid electrolytes combining hierarchical ion-transport prediction algorithms (151–114), *Sci. Data* 7 (1) (2020) 151, <https://doi.org/10.1038/s41597-020-0474-y>.
- [70] Q. Zhao, M. Avdeev, L. Chen, S. Shi, Machine learning prediction of activation energy in cubic Li-argyrodites with hierarchically encoding crystal structure-based (HECS) descriptors, *Sci. Bull. (Beijing)* 66 (14) (2021) 1401–1408, <https://doi.org/10.1016/j.scib.2021.04.029>.
- [71] C. Ling, Phonon-promoted superionic conduction in fluorite-structured compounds, *Chem* 9 (12) (2023) 3588–3599, <https://doi.org/10.1016/j.chempr.2023.08.005>.
- [72] J.-B. Kim, C. Kim, W. Chung, D.H. Yoon, L. Lyu, J. Jang, H.J. Shin, K.-W. Nam, B. Lee, Y.-M. Kang, Anharmonic phonon scattering triggering multi-ion migration in oxide-based superionic conductors, *J. Am. Chem. Soc.* 147 (37) (2025) 33743–33753, <https://doi.org/10.1021/jacs.5c09777>.
- [73] Y. Wang, W.D. Richards, S.P. Ong, L.J. Miara, J.C. Kim, Y. Mo, G. Ceder, Design principles for solid-state lithium superionic conductors, *Nat. Mater.* 14 (10) (2015) 1026–1031, <https://doi.org/10.1038/nmat4369>.
- [74] Y. Zeng, B. Ouyang, J. Liu, Y.W. Byeon, Z. Cai, L.J. Miara, Y. Wang, G. Ceder, High-entropy mechanism to boost ionic conductivity, *Science* 378 (6626) (2022) 1320–1324, <https://doi.org/10.1126/science.abq1346>.
- [75] T. Song, Y. Lin, D. Wang, Q. Chen, C. Ling, S. Shi, Renewing fundamental understanding of ionic transport in inorganic crystalline solid-state electrolytes from the perspective of lattice dynamics, *Adv. Energy Mater.* 14 (34) (2024), <https://doi.org/10.1002/aenm.202302440>.
- [76] M.K. Gupta, J. Ding, N.C. Osti, D.L. Abernathy, W. Arnold, H. Wang, Z. Hood, O. Delaire, Fast Na diffusion and anharmonic phonon dynamics in superionic Na₃PS₄, *Energy Environ. Sci.* 14 (12) (2021) 6554–6563, <https://doi.org/10.1039/d1ee01509e>.
- [77] R. Jaafreh, S. Perezniato, S. Jeong, I.P. Widiartana, J.M. Oh, J.-H. Kang, J. Mun, Y. G. Ko, J.-G. Kim, K. Hamad, et al., Phonon DOS-based machine learning model for designing high-performance solid electrolytes in Li-Ion batteries, *Int. J. Energy Res.* 2024 (1) (2024), <https://doi.org/10.1155/2024/2138847>.
- [78] C. Monroe, J. Newman, The impact of elastic deformation on deposition kinetics at lithium/polymer interfaces, *J. Electrochem. Soc.* 152 (2) (2005) A396, <https://doi.org/10.1149/1.1850854>.
- [79] S. Pugh, Relations between the elastic moduli and the plastic properties of polycrystalline pure metals, *Philos. Mag.* 45 (367) (1954) 823–843, <https://doi.org/10.1080/14786440808520496>.
- [80] J. Mi, J. Yang, L. Chen, W. Cui, Y. Li, X. An, J. Ma, K. Yang, Y. Xie, J. Biao, et al., A ductile solid electrolyte interphase for solid-state batteries, *Nature* 647 (8088) (2025) 86–92, <https://doi.org/10.1038/s41586-025-09675-8>.
- [81] E. Choi, J. Jo, W. Kim, K. Min, Searching for mechanically superior solid-state electrolytes in li-ion batteries via data-driven approaches, *ACS Appl. Mater. Interfaces* 13 (36) (2021) 42590–42597, <https://doi.org/10.1021/acsami.1c07999>.
- [82] J. Sun, S. Kang, J. Kim, K. Min, Accelerated Discovery of Novel Garnet-Type Solid-State Electrolyte Candidates via Machine Learning, *ACS Appl. Mater. Interfaces* 15 (4) (2023) 5049–5057, <https://doi.org/10.1021/acsami.2c15980>.
- [83] Z. Ahmad, T. Xie, C. Maheshwari, J.C. Grossman, V. Viswanathan, Machine learning enabled computational screening of inorganic solid electrolytes for suppression of dendrite formation in lithium metal anodes, *ACS Cent. Sci.* 4 (8) (2018) 996–1006, <https://doi.org/10.1021/acscentsci.8b00229>.
- [84] M.L. Holekevi Chandrappa, J. Qi, C. Chen, S. Banerjee, S.P. Ong, Thermodynamics and Kinetics of the Cathode-Electrolyte Interface in All-Solid-State Li-S Batteries, *J. Am. Chem. Soc.* 144 (39) (2022) 18009–18022, <https://doi.org/10.1021/jacs.2c07482>.
- [85] E.G. Lomeli, B. Ransom, A. Ramdas, D. Jost, B. Moritz, A.D. Sendek, E.J. Reed, T. P. Devereaux, Predicting Reactivity and Passivation of Solid-State Battery Interfaces, *ACS Appl. Mater. Interfaces* 16 (38) (2024) 51584–51594, <https://doi.org/10.1021/acsami.4c06095>.
- [86] K. Kim, N. Adelstein, A. Dive, A. Grieder, S. Kang, B.C. Wood, L.F. Wan, Probing degradation at solid-state battery interfaces using machine-learning interatomic potential, *Energy Stor. Mater.* 73 (2024), <https://doi.org/10.1016/j.ensm.2024.103842>.
- [87] F. Ren, Y. Wu, W. Zuo, W. Zhao, S. Pan, H. Lin, H. Yu, J. Lin, M. Lin, X. Yao, et al., Visualizing the SEI formation between lithium metal and solid-state electrolyte, *Energy Environ. Sci.* 17 (8) (2024) 2743–2752, <https://doi.org/10.1039/d3ee03536k>.
- [88] S. Kumar, R. Jaafreh, S. Dutta, S. Perezniato, K. Hamad, D. Ho Yoon, Learning techniques for designing solid-state lithium-ion batteries with high thermomechanical stability, *Mater. Lett.* 351 (2023), <https://doi.org/10.1016/j.matlet.2023.135049>.
- [89] Y. Liu, Z. Yang, X. Zou, Y. Lin, S. Ma, W. Zuo, Z. Zou, H. Wang, M. Avdeev, S. Shi, A general framework to govern machine learning oriented materials data quality, *Mater. Sci. & Eng. R. Rep.* 166 (2025), <https://doi.org/10.1016/j.mser.2025.101050>.
- [90] F. Yang, E. Campos dos Santos, X. Jia, R. Sato, K. Kisu, Y. Hashimoto, S.-i Orimo, H. Li, A dynamic database of solid-state electrolyte (DDSE) picturing all-solid-state batteries, *Nano Mater. Sci.* 6 (2) (2024) 256–262, <https://doi.org/10.1016/j.nanoms.2023.08.002>.
- [91] S. Wu, H. Li, Evaluation of the application of large language models in the entire process of battery research and development of a comprehensive database for inorganic solid electrolyte, *Acta Phys. Sin.* 74 (16) (2025), <https://doi.org/10.7498/aps.74.20250572>.
- [92] K. Tao, W. He, A. Chen, Y. Han, J. Li, Harnessing lateral transfer learning for pioneering solid electrolyte interphase innovation, *Energy Stor. Mater.* 75 (2025), <https://doi.org/10.1016/j.ensm.2025.104034>.
- [93] Y. Liu, S. Ma, Z. Yang, D. Wu, Y. Zhao, M. Avdeev, S. Shi, Domain knowledge-assisted materials data anomaly detection towards constructing high-performance machine learning models, *J. Mater.* 11 (6) (2025), <https://doi.org/10.1016/j.jmat.2025.101066>.
- [94] T.M. Nguyen, S.A. Tawfik, T. Tran, S. Gupta, S. Rana, S. Venkatesh, The search for superionic solid-state electrolytes using a physics-informed generative model, *Mater. Horiz.* 12 (17) (2025) 6945–6955, <https://doi.org/10.1039/D5MH00767D>.
- [95] A.S. Fuhr, B.G. Sumpter, Deep Generative Models for Materials Discovery and Machine Learning-Accelerated Innovation, *Front. Mater.* 9 (2022), <https://doi.org/10.3389/fmats.2022.865270>.
- [96] Y. Zhao, M. Al-Fahdi, M. Hu, E.M.D. Siriwardane, Y. Song, A. Nasiri, J. Hu, High-throughput discovery of novel cubic crystal materials using deep generative neural networks, *Adv. Sci. (Weinh.)* 8 (20) (2021) e2100566, <https://doi.org/10.1002/advs.202100566>.
- [97] C. Zeni, R. Pinsler, D. Zugner, A. Fowler, M. Horton, X. Fu, Z. Wang, A. Shysheya, J. Crabbe, S. Ueda, et al., A generative model for inorganic materials design, *Nature* 639 (8055) (2025) 624–632, <https://doi.org/10.1038/s41586-025-08628-5>.
- [98] Z. Li, R. Mrad, R. Jiao, G. Huang, J. Shan, S. Chu, Y. Chen, Generative design of crystal structures by point cloud representations and diffusion model, *iScience* 28 (1) (2025) 111659, <https://doi.org/10.1016/j.isci.2024.111659>.
- [99] M. Alverson, S.G. Baird, R. Murdock, S.-H. Ho, J. Johnson, T.D. Sparks, Generative adversarial networks and diffusion models in material discovery, *Digit. Discov.* 3 (1) (2024) 62–80, <https://doi.org/10.1039/d3dd00137g>.